

PREPARATORY PROBLEMS



CUBA NATIONAL CHEMISTRY CONTEST 2022



Ministerio de Educación
República de Cuba

PREPARATORY PROBLEMS

CUBA

National Chemistry Contest 2022

Authors:

BSc. Yadiel Vázquez Mena

BSc. Alvaro Lagar Sosa

Revision:

MSc. Gerardo Manuel Ojeda Carralero

Department of General and Inorganic Chemistry

Faculty of Chemistry

University of Havana

January of 2022

Free distribution

Preface

To the olympists

For the fourth time, we present the Preparatory Problems for the National Chemistry Contest at the high school level. Mentors and students from all over the country have gladly received this booklet. This time, though the name of the people involved in the elaboration change, they are also lecturers at the Faculty of Chemistry from the University of Havana and former contestants in the chemistry olympiads.

It is opportune to highlight how atypical and complex the previous years were, as a result of national and international unexpected events, due to the Covid-19 pandemic. Despite that, we intend to keep providing high-quality material for the preparation for the National Chemistry Contest and the future international olympiads. Unfortunately, and exceptionally, this time, we cannot include an annex section with the results of previous international events. However, we do include the updated Organic Chemistry topics (annex 1). The summary here presented mirrors that from the last booklet; and can be used as a reference and study guide will be very profitable for students and mentors.

As it is not possible in only one contest to encompass all the objectives from the contest program, the problems included here allow to exercise those *Advanced topics*, which are the contents that will be evaluated in the next edition of the national contest (2022) in more depth. In particular, twelfth graders should primarily focus on the stereoisomer of acyclic compounds. It is important to stress that the whole program is still valid. Therefore, the contests from previous years and other kinds of materials discussed with mentors during the national seminars remain valid.

Success in the coming events!

The authors

Havana, January of 2022

Table of contents

Preface	Error! Bookmark not defined.
Table of contents.....	Error! Bookmark not defined.
Constants, conversions and equations	5
Periodic table of the elements.....	5
Proposed problems.....	6
Advanced topics.....	6
Problem 1. Empirical formula	7
Problem 2. Xenon compounds.....	7
Problem 3. Kinetics of NOCl.....	8
Problem 4. Studying a salt.....	9
Problem 5. Multiple choice questions	11
Problem 6. A clay industry	12
Problem 7. Hydrides in Thermochemistry.....	13
Problem 8. Calorimetry	14
Problem 9. Ionic equilibrium	14
Problem 10. Freezing point variation	15
Problem 11. Bromine electrochemistry	16
Problem 12. Identify the compounds.....	16
Problem 13. Insoluble sulfides.....	17
Problem 14. Solutions.....	17
Problem 15. Thujopsene	17
Problem 16. Organic synthesis	19
Problem 17. Study of a drug.....	19
Problem 18. Disidiolide	20
Answers	Error! Bookmark not defined.
Annex 1. Topics of Organic Chemistry for the National Contest	52

CONSTANTS, CONVERSIONS AND EQUATIONS

Avogadro's constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal gas equation	$p \cdot V = n \cdot R \cdot T$
Gases' constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	Thermodynamics First Principle	$Q = \Delta E + W$
Zero in the Celsius scale	273,15 K	Gibbs' energy	$\Delta G = \Delta H - T \cdot \Delta S$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{c_{\text{red}}}{c_{\text{ox}}}$
Ionic product of water at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Arrhenius' equation	$k = A \cdot \exp\left(-\frac{E_A}{R \cdot T}\right)$
Standard conditions	$p^\circ = 100 \text{ kPa}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$		$\Delta G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$
$1 \text{ J} = 1 \text{ C} \cdot 1 \text{ V} = 1 \text{ kPa} \cdot 1 \text{ L} = 1 \text{ W} \cdot 1 \text{ s}$		$1 \text{ C} = 1 \text{ A} \cdot 1 \text{ s}$	

PERIODIC TABLE OF THE ELEMENTS

PERIODIC TABLE OF THE ELEMENTS																	
1 H 1.008																	2 He 4.003
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31	3	4	5	6	7	8	9	10	11	12	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.06	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.63	33 As 74.92	34 Se 78.97	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.95	43 Tc -	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3
55 Cs 132.9	56 Ba 137.3	57-71	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po -	85 At -	86 Rn -
87 Fr -	88 Ra -	89-103	104 Rf -	105 Db -	106 Sg -	107 Bh -	108 Hs -	109 Mt -	110 Ds -	111 Rg -	112 Cn -	113 Nh -	114 Fl -	115 Mc -	116 Lv -	117 Ts -	118 Og -
57 La 138.9	58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm -	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0			
89 Ac -	90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np -	94 Pu -	95 Am -	96 Cm -	97 Bk -	98 Cf -	99 Es -	100 Fm -	101 Md -	102 No -	103 Lr -			

Tabla periódica copiada del examen teórico de la 51 ICHO.

PROPOSED PROBLEMS

Advanced topics

Stoichiometry: Calculations of transformation relations, sample compositions and empirical formulas of organic and inorganic compounds.

Structure: atomic structure; periodic table and periodicity. Types of chemical bonding. Representation of molecules and ions and description of the covalent bonding by Lewis' Formulas, VBT, OMT, VSEPR model; resonance theory and hybridization model.

Descriptive: Balancing equations; ion-electron method (only for 11th and 12th graders). Chemical properties of simple and composed substances derived from the most relevant representative and transition elements; identification of ions/substances. Binary hydrogenated compounds.

Thermochemistry: Enthalpy change. Hess's law. Enthalpies of formation, combustion and bond. Specific heat; calorimetry. Lattice energy; Born-Haber cycle.

Kinetics: Reaction rate. Concentration effect; reaction order and rate law; methods of the initial rates. Activation energy; Arrhenius' equation. Reaction mechanisms; steady-state approximation. Catalysts.

Solutions: Ways to express concentration and composition of samples. Rules to predict electrolytes solubility and strength. Colligative properties of aqueous solutions of (non) electrolytes: vapor pressure, decrease of the freezing point and increase in the boiling point.

Chemical equilibrium: Relations between the thermodynamic properties (T , ΔH , ΔS , ΔG) and the equilibrium constant; applications to the equilibrium in gaseous phase; dissolution-precipitation and in redox processes. Factors influencing the solution-precipitation equilibrium of ionic substances in water: temperature, common ion effect and pH; calculation of the solubility of ionic substances in complex systems.

Electrochemistry: Electrode potential. Galvanic cell; electromotive force. Electrolytic cell; decomposition potential; Faraday's laws. Relations between the thermodynamic properties (ΔH , ΔS , ΔG , ΔE) and the equilibrium constant; Nernst equation.

Organic: Tridimensional conformation of cyclic aliphatic compounds. Condensation reactions in carbonylic compounds; alkene hydroxylation with OsO_4 . Mechanisms of the main organic chemistry reactions. Methods for the identification of carbonylic compounds.

Problem 1. Empirical formula¹

A 0,1152 g sample of a compound formed by carbon, hydrogen, nitrogen and oxygen was burnt in oxygen atmosphere. The gases produced were later treated to convert the nitrogenated products in N_2 . The resultant mixture of CO_2 , H_2O , N_2 and excess O_2 excess, was passed through a desiccant tube filled with $CaCl_2$, whose mass increased by 0,09912 g. The gaseous stream was then bubbled through water in which the CO_2 forms H_2CO_3 . The titration of the solution with aqueous NaOH 0,3283 M requires 28,81 mL to reach the second equivalence point. Excess O_2 was removed by reaction with metallic copper and N_2 was collected in a flask of 225,0 mL volume, giving a pressure of 8,682 kPa at 25 °C. In another experiment, the molar mass of the compound was approximately $150 \text{ g}\cdot\text{mol}^{-1}$.

1.1. Calculate the number of moles of:

- i. H_2O
- ii. CO_2
- iii. N_2

1.2. Determine the mass of the elements in the original compounds:

- i. C
- ii. H
- iii. N
- iv. O

1.3. Find the empirical formula of the compound.

1.4. Find the molecular formula.

Problem 2. Xenon compounds²

Xenon has several applications despite present in the earth atmosphere at trace levels. It is used in for illumination and in optics in flash and arc lamps. Xenon is used as propellant in space shuttles. Also, it has several medicinal applications. Some xenon isotopes are used for the imaging of soft tissues as heart, lung and brain. It is used as general anesthetic and recently its considerable potential for the treatment of cerebral damage due to stroke has been demonstrated.

Xenon is a member of noble gases and has an extremely low reactivity. However, several compounds of xenon with highly electronegative atoms as fluorine and oxygen are known. Xenon reacts with fluorine to form three different xenon fluorides XeF_2 , XeF_4 and XeF_6 . All these fluorides react easily with water, releasing pure gaseous Xe, hydrogen fluoride and molecular oxygen. Xenon oxides and oxyfluorides are obtained by partial or complete hydrolysis of xenon fluorides. Xenon trioxide can be obtained by hydrolysis of XeF_4 or XeF_6 . Hydrolysis of XeF_4 gives XeO_3 , Xe, HF and

¹ National Chemistry Olympiad, United States, 2003. National Exam, Part 2. Problem 1.

² Preparatory Problems for the International Chemistry Olympiad, Turkey 2011. Problem 7.

F₂. However, hydrolysis of XeF₆ only produces XeO₃ and HF. When partially hydrolyzed, XeF₄ and XeF₆ produce XeOF₂ and XeOF₄, respectively, in addition to HF.

2.1. Write balanced chemical equations for the generation of

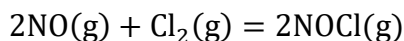
- i. XeO₃ by hydrolysis of XeF₄
- ii. XeO₃ by hydrolysis of XeF₆
- iii. XeOF₂ by partial hydrolysis of XeF₄
- iv. XeOF₄ by partial hydrolysis of XeF₆

2.2. Draw the Lewis structures and tell the hybridization of the central atom of

- i. XeF₂
- ii. XeF₄
- iii. XeO₃
- iv. XeOF₂
- v. XeOF₄

Problem 3. Kinetics of NOCl³

The reaction between nitric oxide and chlorine results in the formation of nitrosyl chloride. At 27 °C the following kinetic data was obtained.

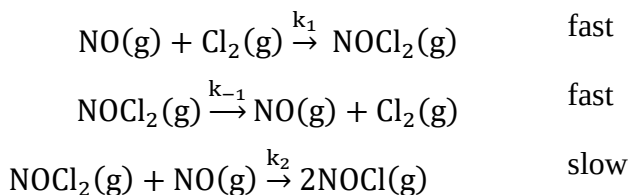


Experiment	Initial concentration (mol·L ⁻¹)		Initial rate (mol·L ⁻¹ ·s ⁻¹)
	NO	Cl ₂	
1	0,020	0,010	8,27·10 ⁻⁵
2	0,020	0,020	1,65·10 ⁻⁴
3	0,020	0,040	3,31·10 ⁻⁴
4	0,040	0,020	6,60·10 ⁻⁴
5	0,010	0,020	4,10·10 ⁻⁵

3.1. Find the rate law for the reaction.

3.2. Calculate the average value for the rate constant.

It is possible that the reaction proceeds according to the mechanism:



³ National Chemistry Contest, Cuba, 2020. Day 1. Common exam. Problem 3

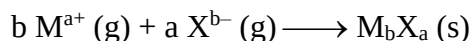
- 3.3. Use the steady-state approximation and find the rate law from the proposed mechanism.
- 3.4. Under what experimental conditions the previous expression (part 3.3) matches the experimental rate law (part 3.1)?
- 3.5. The activation energy of the reaction is known to be $22 \text{ kJ}\cdot\text{mol}^{-1}$. Calculate the rate constant when the temperature increases in 10°C .
- 3.6. Represent the Lewis's structure of NOCl . Classify this molecule according to its polarity.

Problem 4. Studying a salt⁴

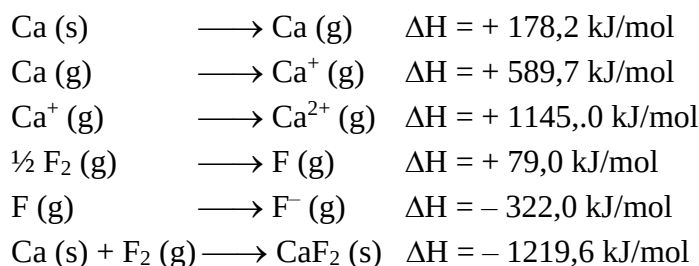
A variety of elements are found in nature in the form of salts. Many of these natural salts are sparingly soluble in water.

On the other hand, the dissolution of some compounds (such as NaOH) in water increases the temperature of the solvent, while the dissolution of others (such as ammonium chloride) cools the media. The solubility and the heat involved during the dissolution depend on the solvation (hydration in the case of water) energy and the lattice energy.

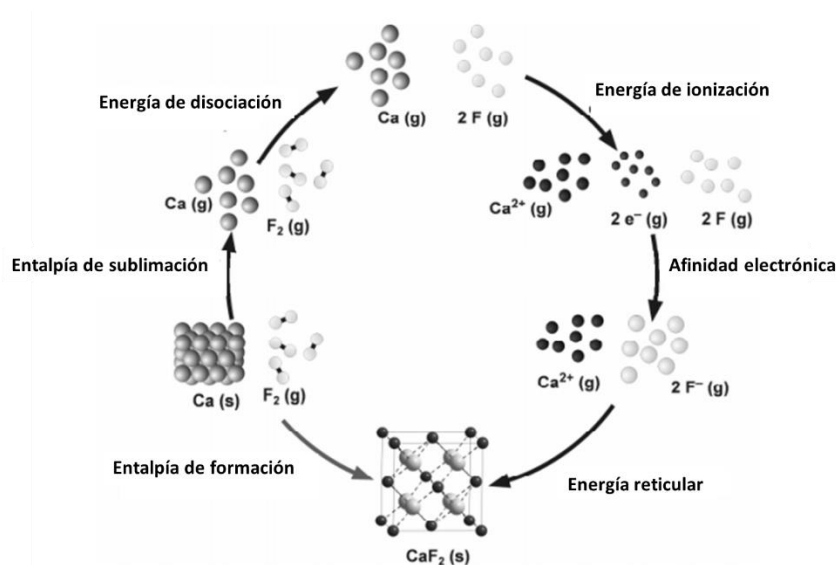
The hydration energy is the amount release due to the interaction of water molecules with the free ions. The lattice energy is the amount released when ions of opposed charges approach to each other –from the infinite– to form a crystal lattice. It provides a measure for the strength of the bonding between the ions in the crystal.



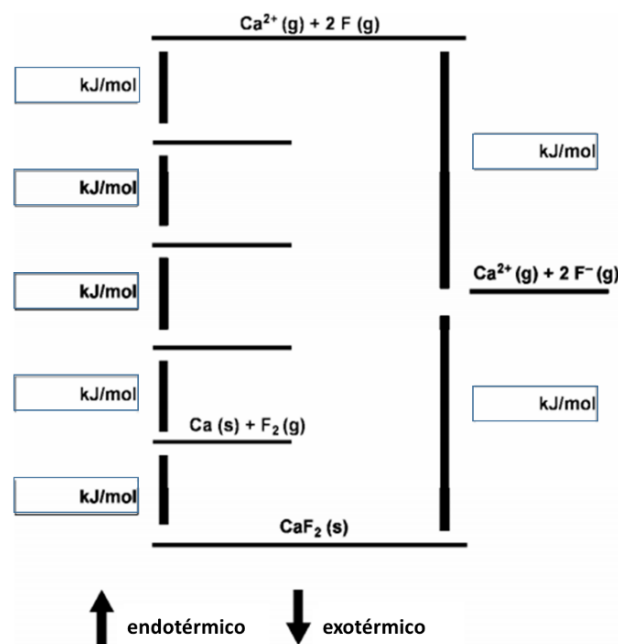
The lattice energy can only be directly determined in a few exceptional cases. For the majority of salts, it can be calculated approximately using the Born-Haber's cycle and another experimentally determined parameters. For this purpose, the energy of the partial steps in the formation of the salt must be considered, and the total process. The formation of calcium fluoride is shown below:



⁴ National Chemistry Olympiad, Germany 2007. Round 1. Problem 1 (*Fragment*)



4.1. Complete the following diagram with the reactions corresponding to the previous steps of the cycle and the associated energy values. Mark endo- and exothermic processes by turning vertical lines into arrows (as indicated below). Calculate the lattice energy of calcium fluoride.



In the nature, calcium fluoride is found in the form of fluorite mineral, which is slightly soluble in water. Generally, the solubility of an ionic compound can be quantified by its solubility product. The solubility product of calcium fluoride is $\text{pK}_{\text{ps}} = 10,52$ at 25°C .

4.2. Write the balance equation for the solubility of calcium fluoride. Calculate the solubility of calcium fluoride in pure water at 25°C , in g/L.

Problem 5. Multiple choice⁵

5.1 Which acid must be stored in plastic flasks instead of glass flasks?

- A) ___ Hydrofluoric acid B) ___ Nitric acid
C) ___ Phosphoric acid D) ___ Sulfuric acid

5.2 In an experiment to prepare bromobenzene according to: $\text{C}_6\text{H}_6 + \text{Br}_2 = \text{C}_6\text{H}_5\text{Br} + \text{HBr}$, a student departed from 20,0 g of C_6H_6 with 0,310 mol de bromo. If 28,0 g of $\text{C}_6\text{H}_5\text{Br}$ were obtained, ¿which is the yield percentage?

- A) ___ 31,5% B) ___ 40,3%
C) ___ 57,6% D) ___ 69,7%

5.3 A 225 mL sample of H_2 is collected over water at 25°C and 735 mmHg. Which expression allow to calculate the dry volume of H_2 at 0°C and 1 atmosphere? The vapor pressure of water at 25°C is 24 mmHg.

- A) ___ $V = \frac{225 \times 760 \times 298}{(735 - 24) \times 273}$ B) ___ $V = \frac{225 \times (735 - 24) \times 273}{760 \times 298}$
C) ___ $V = \frac{225 \times 273 \times 760}{(735 + 24) \times 298}$ D) ___ $V = \frac{225 \times (735 + 24) \times 298}{760 \times 273}$

5.4 Bromine boils at 59 °C with $\Delta H_{\text{vap}} = 29,6 \text{ kJ} \cdot \text{mol}^{-1}$, which is the value for ΔS_{vap} ?

- A) ___ $11,2 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$ B) ___ $89,2 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$
C) ___ $501 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$ D) ___ $1,75 \text{ kJ} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$

5.5 How many unpaired electrons are there in a Co^{3+} gas ion at the ground state?

- A) ___ 1 B) ___ 3
C) ___ 4 D) ___ 5

5.6 Which solution produces a black precipitate when adding an aqueous solution of copper(II)?

- A) ___ NH_3 B) ___ $(\text{NH}_4)_2\text{S}$
C) ___ K_2SO_4 D) ___ NaOH

5.7 Which oxide is the best reducing reagent?

- A) ___ CO_2 B) ___ NO_2
C) ___ SiO_2 D) ___ SO_2

5.8 Methylamine, CH_3NH_2 , reacts with oxygen to form CO_2 , N_2 and H_2O . Which amount of oxygen (in moles) is needed to react completely with 1,00 mol of methylamine?

- A) ___ 2,25 B) ___ 2,50
C) ___ 3,00 D) ___ 4,50

5.9 Toluene, C_7H_8 , is added to gas to increase the octane-grade. Which is the ration air volume to toluene(g) for the complete combustion of toluene? (Assume that the air is composed of 20% O_2)

- A) ___ 9/1 B) ___ 11/1
C) ___ 28/1 D) ___ 45/1

⁵ National Chemistry Olympiad, United States, 2006. National Exam, Part 1. Fragments.

5.10 Which of the following elements has the highest second ionization energy?

A) ___ Na

B) ___ Mg

C) ___ Al

D) ___ K

Problem 6. A clay industry⁶

In the socioeconomic environment where Universitat Jaime I is place, the ceramic sector has great importance, in the municipality of Castellón more than 90% of the manufacture of clay bridge is concentrated. One auxiliar sector is the manufacture of enamel and ceramic pigments. These processes are carried out at temperatures higher than 1000 °C.

Seeger's formula is used to express the composition of ceramic enamels. For the calculation, the oxides are classified in basic (M_2O , MO), amphoteric (M_2O_3) and acid (MO_2) and the substance amount of each kind of oxide is indicated taking at reference one mole of total basic oxides. Oxide B_2O_3 is an exception, as it is considered acid. For example, Seeger's formula (molar ratio of amphoteric and acid oxides, with respect to basic oxides) for a green enamel is: $(M_2O, MO) \cdot 0,455 (M_2O_3) \cdot 3,243 (MO_2)$.

6.1 Find the Seeger's formula for a ceramic enamel that contains 85% of "frite" (oxide mixture) and 15% of kaolin ($Al_2O_3 \cdot 2SiO_2 \cdot 2H_2O$). Do not take into account volatile oxides that might be present.

<u>Raw material for frite</u>	<u>Mass percentage</u>
-------------------------------	------------------------

Feldspar ($K_2O \cdot Al_2O_3 \cdot 6SiO_2$)	60%
--	-----

Borax ($Na_2B_4O_7 \cdot 10H_2O$)	20%
-------------------------------------	-----

Creta ($CaCO_3$)	10%
--------------------	-----

Quartz (SiO_2)	7%
--------------------	----

Boric acid (H_3BO_3)	3%
--------------------------	----

Indicate the amount of substance of each component and the amount of oxide to obtain 100 g of enamel.

6.2 If you have the following raw material in the laboratory (purity is expressed in mass percentage):

Y_2O_3 (99,9 %), Al_2O_3 (99,5 %), Nd_2O_3 (99,8 %), Cr_2O_3 (99 %), Co_3O_4 (98 %).

Calculate the amount needed to obtain the following ceramic pigments:

a) 10,00 g of $Y_{1-x}Nd_xAl_{1-x}Cr_xO_3$ where $x = 0,05$

⁶ XIX Iberoamerican Chemistry Olympiad, Spain, 2004. Question 2.

b) 10,00 g of $\text{CoAl}_{1,25}\text{Cr}_{0,75}\text{O}_4$

Molar masses (g/mol):

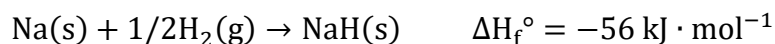
$\text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2 \cdot 2\text{H}_2\text{O}$	$\text{K}_2\text{O} \cdot \text{Al}_2\text{O}_3 \cdot 6\text{SiO}_2$	$\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$	$\text{Y}_3\text{Al}_5\text{O}_{12}$
258,158	556,670	381,367	593,611

$\text{Y}_{0,95}\text{Nd}_{0,05}\text{Al}_{0,95}\text{Cr}_{0,05}\text{O}_3$	$\text{CoAl}_{1,25}\text{Cr}_{0,75}\text{O}_4$
167,901	195,653

CaCO_3	SiO_2	H_3BO_3	Y_2O_3	Al_2O_3	Nd_2O_3	Cr_2O_3	Co_3O_4
100,088	60,084	61,832	225,807	101,960	336,477	151,989	240,795

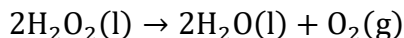
Problem 7. Thermochemistry hydrides⁷

Sodium hydride is formed by the reaction between hydrogen and sodium.



7.1. Calculate the lattice energy of this substance using the thermodynamic cycle of Born-Haber.

Hydrogen peroxide is thermodynamically unstable and decomposes slowly forming water and oxygen. This process is accelerated in the presence of bases.



7.2. Given the values of mean dissociation enthalpy, calculate approximately the enthalpy variation for the previous reaction.

Methane is the major component of natural gas. This hydrocarbon is produced by the putrefaction of organic matter, naturally or due to human action. Its main application is as fuel.

7.3. Calculate the combustion enthalpy of methane, this is, the enthalpy variation corresponding to the reaction of methane with oxygen.

Part of the methane is found in the bottom of the sea in the form of combustible ice, which is formed by a clathrate of the ideal composition $(\text{CH}_4)_8(\text{H}_2\text{O})_{46}$. This methane hydrates are formed at high pressure and low temperature.

7.4. Which amount of energy can be obtained by the combustion of 1,00 g of methane hydrate?

One of the global warming dangers is that it may trigger the release of methane stored in the bottom of the ocean and in the permafrost. Methane is a gas with a green effect stronger than CO_2 , therefore

⁷ National Chemistry Contest, Cuba 2019. Day 2. 10th grade. Problem 7

its release in huge amounts to the atmosphere would accelerate further the global warming. It is estimated that the total amount of methane in the Earth is larger than $5 \cdot 10^{11}$ tons.

7.5. How many degrees would the earth atmosphere be heated if that amount of methane is burnt with the atmospheric oxygen? The total heat capacity of the earth atmosphere has an approximate value of $4 \cdot 10^{21} \text{ J} \cdot \text{K}^{-1}$.

$I(\text{H}) = 1312 \text{ kJ/mol}$, $I(\text{Na}) = 496 \text{ kJ/mol}$, $A(\text{H}) = -73 \text{ kJ/mol}$, $A(\text{Na}) = -53 \text{ kJ/mol}$, $\Delta H_{\text{dis}}^\circ(\text{H}_2) = 436 \text{ kJ/mol}$, $\Delta H_{\text{dis}}^\circ(\text{Na}_2) = 77 \text{ kJ/mol}$, $\Delta H_{\text{sub}}^\circ(\text{Na}) = 107 \text{ kJ/mol}$, $\Delta H_{\text{f}}^\circ(\text{NaH}) = -56 \text{ kJ/mol}$

Bond	H-O	O-O	O=O
$E_{\text{dis}}/\text{kJ} \cdot \text{mol}^{-1}$	463	146	497

Substance	CH_4	H_2O	CO_2
$\Delta H_{\text{f}}^\circ/\text{kJ} \cdot \text{mol}^{-1}$	-75	-286	-394

Problem 8. Calorimetry⁸

It is possible to use calorimetry experiments in a coffee pot to obtain the enthalpy of formation ($\Delta H_{\text{f}}^\circ$) for magnesium oxide.

8.1 Write the balanced equation for the formation of magnesium oxide, with an enthalpy change equal to $\Delta H_{\text{f}}^\circ$.

8.2 To determine the heat capacity of the calorimeter, 49,6 mL of HCl 1,01 M are reacted to 50,1 mL of NaOH 0,998 M. The dissolution temperature increases in 6,40 °C. Determine the heat capacity of the calorimeter. You may assume that the specific heat capacity of the solution is $4,025 \text{ J} \cdot \text{g}^{-1} \cdot ^\circ\text{C}^{-1}$ and the density is 1 g/mL, and the enthalpy of neutralization is -55,9 kJ per mole of H_2O .

8.3 When 0,221 g of magnesium chips are added to 49,9 mL of HCl 1,01 M and 49,7 mL of H_2O in the same calorimeter, the temperature increases by 9,67 °C. Write a balanced equations for the reaction that happen and calculate ΔH per mol of magnesium. (Suppose that the specific heat capacity of the solution is $3,862 \text{ J} \cdot \text{g}^{-1} \cdot ^\circ\text{C}^{-1}$ and the calorimeter constant is the value obtained in **8.2**)

8.4 When 0,576 g of MgO reacts with 51,0 mL of HCl 1,01 M and 50,1 mL of H_2O in the same calorimeter the temperature increases 4,72 °C. Write the balanced equation for this reaction and calculate its ΔH per mol of MgO using the same approaches than in **8.3**.

8.5 Use the previous results and $\Delta H_{\text{f}}^\circ$ for $\text{H}_2\text{O} (\text{l})$ ($-285,8 \text{ kJ} \cdot \text{mol}^{-1}$) to calculate $\Delta H_{\text{f}}^\circ$ for magnesium oxide.

Problem 9. Ionic equilibrium⁹

In the following, there are several examples of frequently employed calculations for the quantitative treatment of the different electrolytes equilibria that are established in aqueous solutions.

⁸ National Chemistry Olympiad, United States, 2007. National Exam, Part 2. Problem 2.

⁹ National Chemistry Contest, Cuba, 2019. Day 2. 12th grade. Problem 10.

9.1 Which is the volume of hydrochloric acid at 37% in mass ($\rho_D = 1,19 \text{ g}\cdot\text{mL}^{-1}$) needed to prepare 500 mL of a solution with an approximate concentration of $0,50 \text{ mol}\cdot\text{L}^{-1}$?

9.2 Is there any precipitate when gaseous hydrogen sulfide saturates an acid solution of pH equal to 3 that contains Pb^{2+} ions at a concentration of $5,0\cdot 10^{-5} \text{ mol}\cdot\text{L}^{-1}$? It is known that in these conditions the total concentration of sulfide is around $0,10 \text{ mol}\cdot\text{L}^{-1}$.

$K_a(\text{H}_2\text{S}) = 1,0\cdot 10^{-7}$; $K_a(\text{HS}^-) = 1,3\cdot 10^{-13}$; $K_{ps}(\text{PbS}) = 2,5\cdot 10^{-27}$

9.3 Calculate the solubility of silver sulfate in water in $\text{mol}\cdot\text{L}^{-1}$. $K_{ps}(\text{Ag}_2\text{SO}_4) = 1,2\cdot 10^{-5}$

9.4 Calculate the pH of the resultant dissolution after mixing 10 mL of hydrochloric acid $0,60 \text{ mol}\cdot\text{L}^{-1}$ with 20 mL of ammonia $0,50 \text{ mol}\cdot\text{L}^{-1}$. $K_b(\text{NH}_3) = 1,8\cdot 10^{-5}$

Problem 10. Freezing point variation¹⁰

In a course project, a student is planning to make tests with aqueous solutions of different solutes to estimate their properties.

To carry out the experiments, two solutes **A** and **B** (non-electrolytes) were given to the student. The mass of **A** is 180 g/mol, while the mass of **B** is 342 g/mol.

Your task is to help this student with some of the experiment. Consider that the ebullioscopic constant for water is $K_b = 0,51 \text{ K}\cdot\text{kg/mol}$ and the cryoscopic constant is $K_c = 1,86 \text{ K}\cdot\text{kg/mol}$.

10.1 Determine the melting point (freezing point) of a solution containing 30 g of solute **A** in 100 g of water.

10.2 A commercial refrigerator works at a temperature of 7°C below zero. Calculate the maximum mass of solute **A** in grams that can be added to 5000 g of water to guarantee that the refrigerator freezes this solution.

10.3 Our student friend was given 1 kg of a solution of substance **C**, containing 70% mass of water, to study its properties. After performing cryoscopy test, the student found that the freezing point of the solution is $3,03^\circ\text{C}$ below zero. Determine the molecular mass of the solute.

10.4 In an additional test, the student was given another kilogram of a solution similar to the previous one. After the cryoscopy test, the student also found that the freezing point was $3,03^\circ\text{C}$ below zero, however in this case there were 300 g of a mixture of solids **A** and **B** in 700 g of water. Calculate the molality of **A** in the previous solution.

10.5 Finally, our friend was given a solution containing equal masses of solutes **A** and **B** dissolved in 400 g of water. The determination of the freezing point of this solution rendered $2,5^\circ\text{C}$ below zero. Calculate the mass of **A** (in grams) in the solution.

¹⁰ XV Iberoamerican Chemistry Olympiad, Mexico, 2010. Theoretical exam. Problem 1.

Problem 11. Bromine electrochemistry¹¹

Many inorganic compounds cannot be obtained by direct methods because of the occurrence of unwanted side processes, for example, oxyacids and oxysalts of halogens. The following diagram shows the potentials of bromine species in basic media ($[\text{OH}^-] = 1,00 \text{ mol L}^{-1}$) and the standard electrode reduction potentials in volts associated to each process:



11.1 Predict whether the species BrO^- is thermodynamically unstable with respect to its disproportionation to BrO_3^- and Br_2 species and justify your answer. (Write the redox equation and calculate the potential)

11.2 Write the value of $E^\circ \text{Br}_2/\text{Br}^-$ in basic media.

11.3 Determine the reduction potential of bromate ion to bromide.

Problem 12. Identify the compounds¹²

Organic compound **A** has 41,38% carbon, 3,45% hydrogen and the rest is oxygen. When heating compound **A** with ethanol in the presence of an acid, a new substance **B** is obtained, containing 55,81% carbon, 6,97% hydrogen and the rest in oxygen.

Initial compound **A** reacts with hydrobromic acid to give product **C**, which reacts with boiling water to give substance **D**, constituted by 35,82% carbon, 4,48% hydrogen, besides oxygen. 2,68 g of substance **D** are needed to react completely with 20 cm³ of potassium hydroxide 2 N.

12.1 Determine the structural formula of **A**, **B**, **C** and **D**. Take into account that compound **A** releases water when heated.

12.2 Write the chemical equations corresponding to the previous processes.

¹¹ V Iberoamerican Chemistry Olympiad, Spain, 1999. Problem 2, exercise IV.

¹² IV International Chemistry Olympiad, Russian, 1972. Theoretical Exam. Problem 6.

Problem 13. Insoluble sulfides¹³

Many metallic sulfides are known to be insoluble in aqueous media.

13.1 Demonstrate that 0,10 mol of Ti_2S is dissolved in a liter of a 1,0M solution of any non-coordinated monoprotic strong acid.

13.2 Demonstrate that 0,10 mol of CuS is dissolved in 1,0 L of 1,0 M HNO_3 but not in 1,0 L of 1,0 M HCl .

Assume that Cu^{2+} ions do not form stable complexes with chloride ions.

Data:

$$E_0(\text{S}/\text{S}^{2-}) = -0,48 \text{ V} \quad E_0(\text{NO}_3^- / \text{NO}(\text{ac})) = 0,96 \text{ V}$$

$$\text{p}K_a(\text{H}_2\text{S}) = 7,0 \quad \text{p}K_a(\text{HS}^-) = 13,0$$

$$K_{\text{ps}}(\text{Ti}_2\text{S}) = 1 \cdot 10^{-20} \quad K_{\text{ps}}(\text{CuS}) = 1 \cdot 10^{-35}$$

$$\text{Solubility of NO in water (298 K): } 2,53 \cdot 10^{-2} \text{ mol} \cdot \text{dm}^{-3}$$

$$\text{Solubility of H}_2\text{S in water (298 K): } 0,1 \text{ mol} \cdot \text{dm}^{-3}$$

Problem 14. Solutions¹⁴

A solution containing 160,0 g of water and 100,0 g of calcium nitrate is electrolyzed between graphite electrodes for 12 h with an electric current of 5,00 A. At the end of the electrolysis, the mass of the solution diminished by 41,9 g.

14.1 Determine the amount of tetra-hydrated calcium nitrate ($\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$) that can be dissolved in 100,0 g of water at this temperature.

The solubility of sparingly soluble salts can be influenced by two important factors, one of which is the pH. The solubility of silver oxalate is $S = 2,06 \cdot 10^{-4} \text{ mol/L}$.

14.2 Explain why pH influences the solubility of silver oxalate.

14.3 Calculate the solubility of silver oxalate at $\text{pH} = 5$.

Data:

$$K_{a1}(\text{H}_2\text{C}_2\text{O}_4) = 5,9 \cdot 10^{-2} \text{ mol/L} \quad K_{a2}(\text{H}_2\text{C}_2\text{O}_4) = 6,4 \cdot 10^{-5} \text{ mol/L}$$

Problem 15. Thujopsene¹⁵

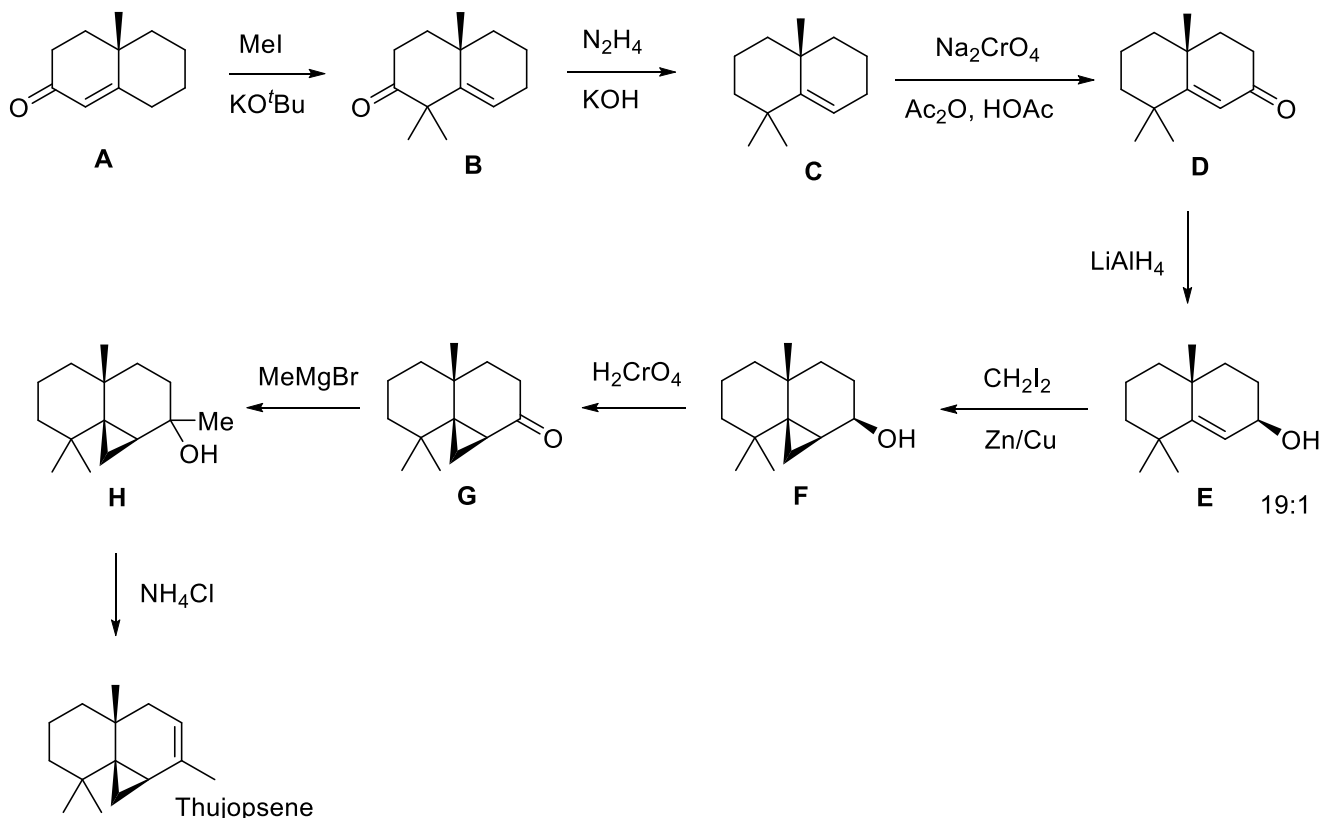
Thujopsene is a chemical natural compound, it is classified as a sesquiterpene, with a molecular formula of $\text{C}_{15}\text{H}_{24}$. Thujopsene I is found in the essential oil of a kind of conifer, in particular, *Juniperus cedrus* and *Thujopsis dolabrata*, representing around 2,2% of weight of the duramen, the

¹³ XXIII International Chemistry Olympiad, Polonia, 1991. Theoretical Exam. Problem 1. (Adaptation)

¹⁴ German Chemistry Olympiad, 2003. Round 2. Problem 2-3. (Adaptation).

¹⁵ National Chemistry Contest, Cuba 2020. Preparatory Problems. Problem 23.

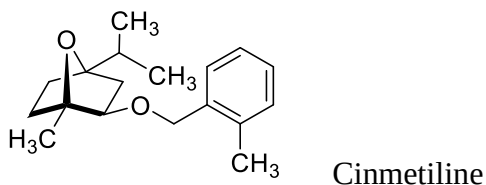
woody part of the plant trunk. This compound can be obtained according to the following synthetic sequence (Dauben, W. G.; Ashcraft, A. C., *J. A. Chem. Soc.* **1963**, 85, 3673).



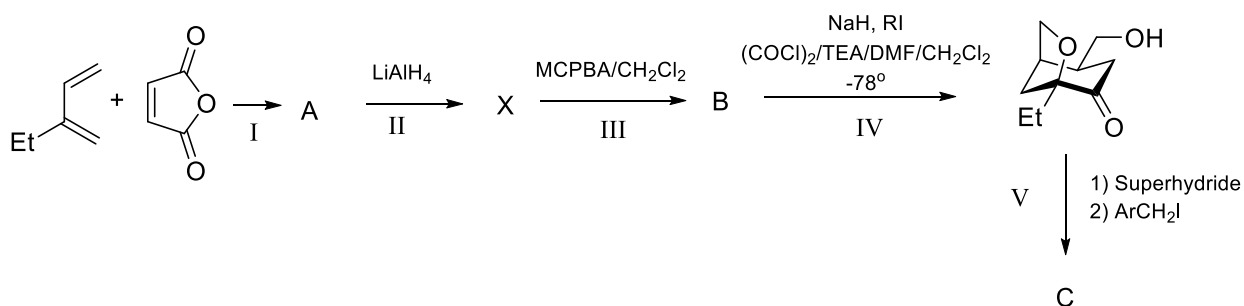
- 15.1 How many stereoisomers are possible for thujopsene?
- 15.2 Assign the absolute configuration of all the asymmetric carbons of **F**.
- 15.3 Why is alcohol **E** the major stereoisomeric product of the reduction of **D**?
- 15.4 Why is the formation of **F** stereoselective?
- 15.5 Propose a plausible mechanism for each of the chemical reactions of the sequence starting from **B**.
- 15.6 Propose a way to synthesize the starting material **A**, from cyclohexanone and methylvinylketone.

Problem 16. Organic synthesis

Using raw material of easy access an oxabicyclic ether can be obtained, with herbicide properties, that can be considered as a mimetic (similar structure) of Cinmetiline. (Romero-Martínez *et al.* *Revista de la Sociedad Química de México*, **2002**, 46 (1), p. 54-63)



The following scheme shows the synthetic route that was employed:

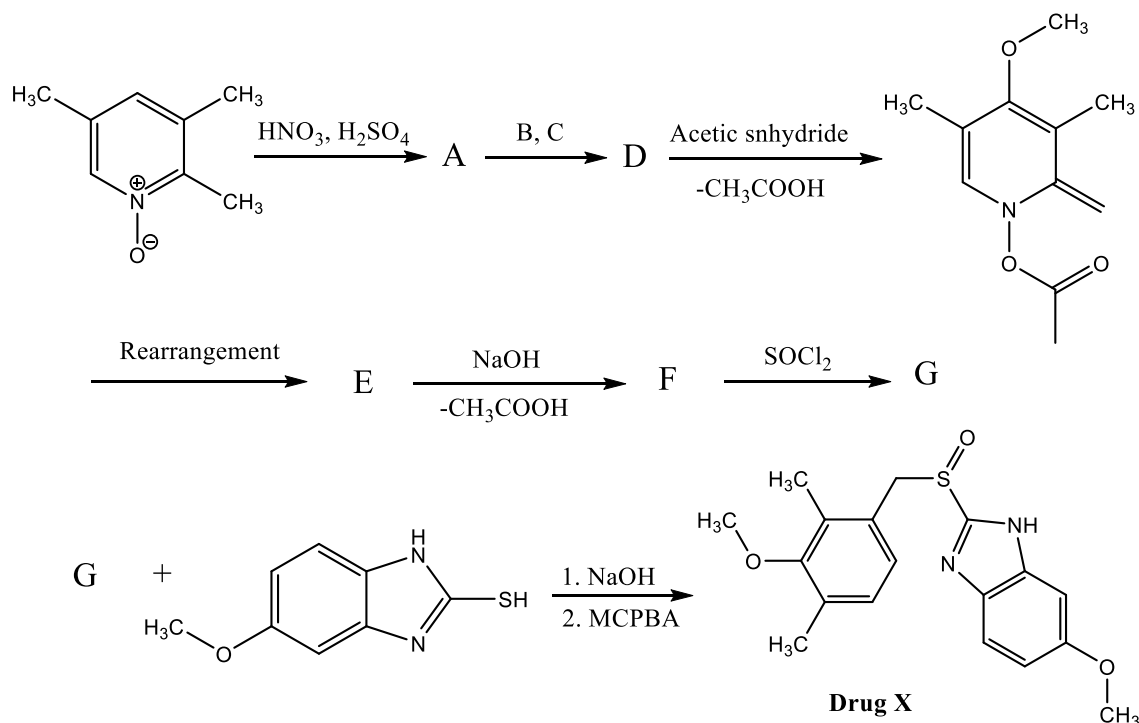


- 16.1** Which stereoisomer of **A** can be obtained in the reaction **I**, which is the major product and why?
- 16.2** Represent the chair structure of **B** taking into account that in reaction **III** there is a selective oxidation to form the free carbonyl group.
- 16.3** In reaction **II**, after the reduction with LiAlH_4 , an intermediate compound **X** is formed, that after treatment with MCPBA forms **B**. Represent the structure of **X**.
- 16.4** Propose a mechanism for the transformation of **X** into **B**, showing always the stereochemistry of the plausible intermediates.
- 16.5** Draw the chair structure of **C** and assign the absolute configuration of the assimetric carbons.

Problem 17. Study of a drug¹⁶

Below the medicine **X** is shown. This inhibits the proton release and suppress the secretion of hydrochloric acid in the stomach. It is used for the treatment of the gastric ulcer, among others. The synthesis of compound **X** is carried out though the following route:

¹⁶ German Chemistry Olympiad, 2007. Round 2. Problem 2-3. Adaptation.



17.1 Write the name of the starting compounds. Draw the structures of the compounds A, D, E, F and G.

17.2 Propose a mechanism for the reaction of D with acetic anhydride. (Take into account the tautomerism)

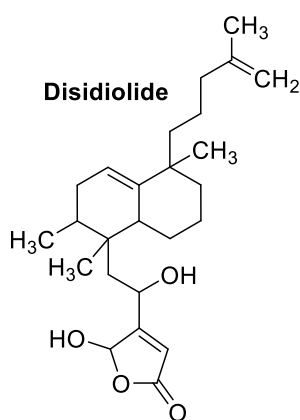
17.3 Which are the reagents B and C used in the transformation from A to D? What is the name of this reaction?

17.4 Propose a mechanism for the rearrangement. Which is the driving force of this reaction?

17.5 Why is MCPBA (*m*-chloro-peroxybenzoic acid) used in the last step of the synthesis?

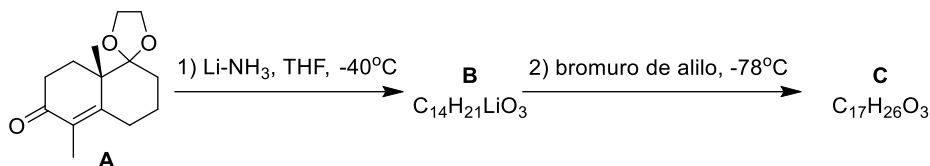
17.6 Is the final product of the synthesis chiral? Argument your answer.

Problem 18. Disidiolide



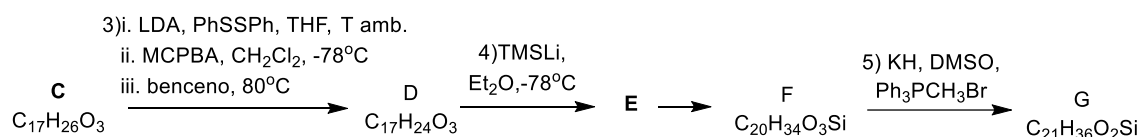
Disidiolide, a compound isolated from the Caribbean sponge *Dysidea etheria* from Laubenfels is considered the first inhibitor of the human phosphatase protein cdc25A. That inhibition ability blocks the cyclins-depending dephosphorylation of kinases processes, which triggers apoptosis of human cancer cells. **Disidiolide** is a γ -hydroxybutenolide derivative (α,β -unsaturated lactone) of a class of terpenes that present in its backbone the [4.4.0]bicyclic moiety and several stereocenters. Such structural features make very attractive the total synthesis, which will be described in detail in the following discussion.

First, under Birch reduction conditions compound **B** is obtained starting from **A**, and then compound **B** is alkylated at low temperatures with allyl bromide to form exclusively compound **C**.



18.1.1 Represent the tridimensional structure of compounds **B** and **C** employing the chair conformation. Write the mechanism of step 1 and explain why only the diastereoisomers **B** and **C** are formed in each synthetic step, respectively.

Compound **C** is converted to the correspondent α,β -enone **D** by a sequence of sulfonylation-dehydrosulfonylation. Next, the fast addition of excess trimethylsilyl lithium produces an intermediate compound **E** that in the presence of water allows the formation of compound **F**. Consider that the regioselectivity of the trimethylsilyl lithium is opposed to the organolithium reagents when reacting with α,β -enones. Then, the reaction of **F** with methylenetriphenylphosphorane bromide producing compound **G** with excellent yields.

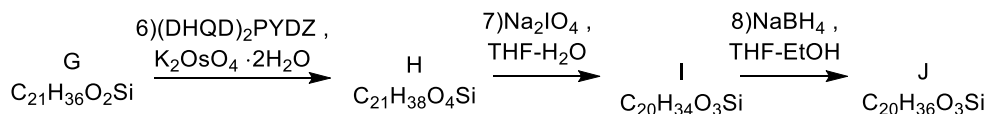


18.2.1 Represent the tridimensional structure of compounds **D**, **E**, **F** and **G**.

18.2.2 Write the tridimensional structure of the compounds **D** and **E** employing the chair conformation. Explain why in the reaction 4 diastereoisomer **E** is preferentially formed.

18.2.3 What is the name of reaction 5? Write the mechanism of the reaction.

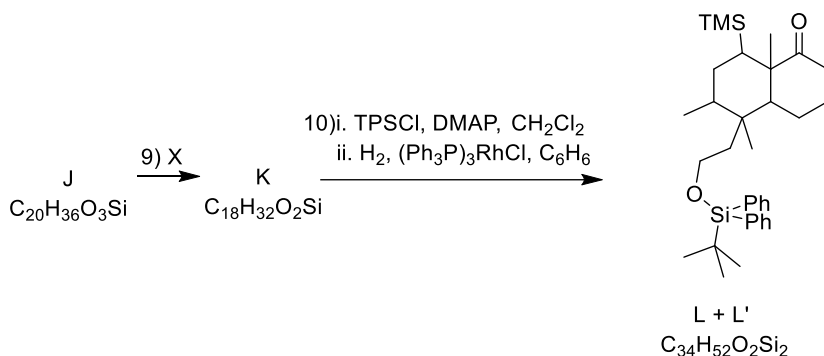
Compound **G** is subjected to a modern variation of the classic dihydroxylation with osmium tetroxide, denominated Sharpless dihydroxylation. In this reaction, a hindered ligand (DHQD)₂PYDZ (pyridazine bonded to two dehydroquinidines) is added under the typical conditions of dihydroxylation to obtain compound **H**. Compound **H** is treated with sodium periodate producing compound **I**, which is reactive towards 1 mol of 2,4-dinitrophenylhydrazine and with Tollen's test produces a silver mirror. Finally, **I** is reacted with sodium borohydride to get compound **J**.



18.3.1 Represent the tridimensional structure of compounds **H**, **I** and **J**.

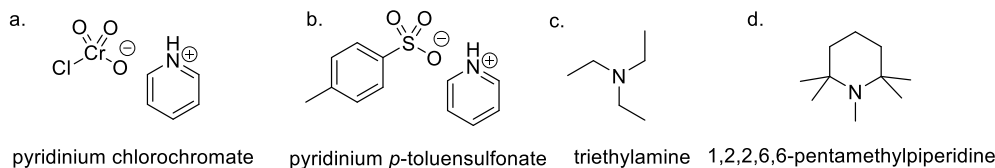
18.3.2 Sharpless' dihydroxylation employs conditions that considerably reduce the amount needed to osmium tetroxide, which is a reagent highly toxic and costly. Which other advantage of Sharpless' dihydroxylation with respect to the classic reaction was taken in the synthetic route?

By acid catalyzed hydrolysis, the acetal group is removed from compound **J** to give **K**. Then, hydroxyl group from **K** was protected generating a silyl ether of *tert*-butyldiphenylsilyl (TPSCI) kind and next, it reacts with H_2 at high pressures in the presence of Wilkinson's catalyst to give a mixture of diastereoisomers **L** and **L'**. Compound **L** is the least stable of both diastereoisomer obtained and is later isolated in its diasteromerically pure form by a chromatographic separation.



18.4.1 Represent the tridimensional structure of compounds **K**, **L** and **L'**.

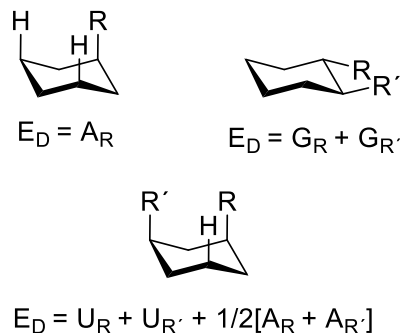
18.4.2 Among the following compounds, choose which reagent **X** can be employed in step 9.



18.4.3 Calculate the difference of the destabilization energy between diastereoisomers **L** and **L'**. Employ the data in the following table for the destabilization energy (E_D) by 1,3-diaxial interactions between a hydrogen atom and substituent **R** (A_R), gauche interactions and 1,3-diaxial interactions between substituents **R** and **R'** (G_R and U_R , respectively).

Table 18.4.3 A_R , G_R and U_R values for the cyclohexane derivatives (kcal/mol).

R o R'	A_R	G_R	U_R
CH ₃	1.8	0.9	1.8
CH ₂ R	1.8	0.4	1.8
SiR ₃	~6.0	~2.5	~6.0
OH	0.9	0.2	0.9

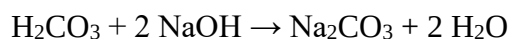


Compound **L** was reacted with allylmagnesium bromide rendering stereoselectively **M**. Next, compound **M** is reacted with dimethylsulfideborane, followed by a peroxide oxidation to give diol **N**. Finally, the least sterically hindered hydroxyl group from compound **N** is selectively protected through the formation of a *tert*-butyldimethylsilyl ether (TBS) to give compound **O**.

ANSWERS

Problem 1

- The sample contains 0,1152 g including C, H, N and O.
- 0,0912 g of H₂O were formed.
- Conversion of carbon dioxide to carbonic acid has 1 to 1 stoichiometry.
- The total neutralization of carbonic acid is achieved with 28,81 mL of NaOH 0,3283 M according to:



- The volume of N₂ collected is 225,0 mL at 8,682 kPa and 25°C.
- The molar mass of the substance is approximately 150 g/mol

1.1 i. H₂O

$$n(\text{H}_2\text{O}) = \frac{m(\text{H}_2\text{O})}{M(\text{H}_2\text{O})} = \frac{0,09912 \text{ g}}{18,02 \text{ g} \cdot \text{mol}^{-1}} = 5,501 \cdot 10^{-3} \text{ mol}$$

ii. CO₂

$$n(\text{CO}_2) = 0,5 * n(\text{NaOH}) = 0,5 * c(\text{NaOH}) * V_{\text{val.}}(\text{NaOH})$$

$$n(\text{CO}_2) = 0,5 * 0,3283 \text{ mol} \cdot \text{L}^{-1} * 0,02881 \text{ L} = 4,729 \cdot 10^{-3} \text{ mol}$$

iii. N₂

$$n(\text{N}_2) = \frac{p(\text{N}_2) * V(\text{N}_2)}{R * T} = \frac{8,682 \text{ kPa} * 0,225 \text{ L}}{8,3145 \text{ kPa} \cdot \text{L} \cdot \text{K}^{-1} \cdot \text{mol}^{-1} * 298,15 \text{ K}} = 7,88 \cdot 10^{-4} \text{ mol}$$

1.2 i. C

$$m(\text{C}) = n(\text{CO}_2) * M(\text{C}) = 4,729 \cdot 10^{-3} \text{ mol} * 12,01 \text{ g} \cdot \text{mol}^{-1} = 0,05680 \text{ g}$$

ii. H

$$m(\text{H}) = 2 * n(\text{H}_2\text{O}) * M(\text{H}) = 2 * 5,501 \cdot 10^{-3} \text{ mol} * 1,008 \text{ g} \cdot \text{mol}^{-1} = 0,01109 \text{ g}$$

iii. N

$$m(\text{N}) = 2 * n(\text{N}_2) * M(\text{N}) = 2 * 7,88 \cdot 10^{-4} \text{ mol} * 14,01 \text{ g} \cdot \text{mol}^{-1} = 0,02208 \text{ g}$$

iv. O

$$m(O) = m(\text{muestra}) - (m(C) + m(H) + m(N))$$

$$m(O) = 0,1152 \text{ g} - (0,05680 \text{ g} + 0,01109 \text{ g} + 0,02208 \text{ g}) = 0,02523 \text{ g}$$

1.3 Empirical formula

$$C: 0,05680 \text{ g} \div 12,01 \text{ g} \cdot \text{mol}^{-1} = 4,729 \cdot 10^{-3} \text{ mol} \div 1,576 \cdot 10^{-3} \text{ mol} = 3,001 \approx 3$$

$$H: 0,01109 \text{ g} \div 1,008 \text{ g} \cdot \text{mol}^{-1} = 1,100 \cdot 10^{-2} \text{ mol} \div 1,576 \cdot 10^{-3} \text{ mol} = 6,980 \approx 7$$

$$N: 0,02208 \text{ g} \div 14,01 \text{ g} \cdot \text{mol}^{-1} = 1,576 \cdot 10^{-3} \text{ mol} \div 1,576 \cdot 10^{-3} \text{ mol} = 1$$

$$O: 0,02523 \text{ g} \div 16,00 \text{ g} \cdot \text{mol}^{-1} = 1,577 \cdot 10^{-3} \text{ mol} \div 1,576 \cdot 10^{-3} \text{ mol} = 1,001 \approx 1$$

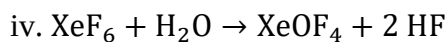
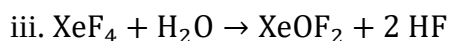
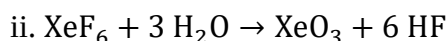
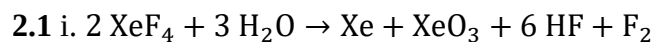
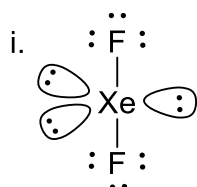
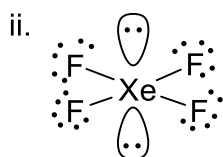
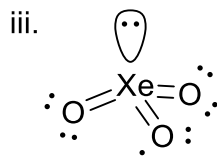
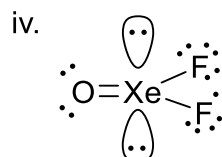
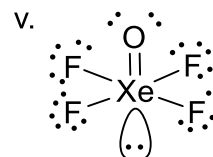
Therefore, the empirical formula of the compound is C_3H_7NO .

1.4 Molecular formula

The molar mass of the empirical formula is $73,1 \text{ g} \cdot \text{mol}^{-1}$. When comparing the empirical mass with the molar mass, one gets:

$$150 \text{ g} \cdot \text{mol}^{-1} \div 73,1 \text{ g} \cdot \text{mol}^{-1} = 2,05 \approx 2$$

Then, the molecular formula is $C_6H_{14}N_2O_2$.

Problem 2**2.2** sp^3d  sp^3d^2  sp^3  sp^3d  sp^3d^2

Problem 3**3.1 Rate law**

$$v = k_r \cdot c^a(\text{NO}) \cdot c^b(\text{Cl}_2)$$

$$\frac{\text{Exp 1}}{\text{Exp 2}} = \frac{8,27 \cdot 10^{-5}}{1,65 \cdot 10^{-4}} = \frac{k_r \cdot 0,020^a \cdot 0,010^b}{k_r \cdot 0,020^a \cdot 0,020^b}$$

$$0,501 = 0,5^b \rightarrow b = 1$$

$$\frac{\text{Exp 2}}{\text{Exp 4}} = \frac{1,65 \cdot 10^{-4}}{6,60 \cdot 10^{-4}} = \frac{k \cdot 0,020^a \cdot 0,020^b}{k \cdot 0,040^a \cdot 0,020^b}$$

$$0,25 = 0,5^a \rightarrow a = 2$$

$$v = k_r \cdot c^2(\text{NO}) \cdot c(\text{Cl}_2)$$

3.2 Rate constant

$$k_r = \frac{v}{c^2(\text{NO}) \cdot c(\text{Cl}_2)}$$

$$\text{Exp 1} = 20,675 \quad \text{Exp 2} = 20,625 \quad \text{Exp 3} = 20,6875$$

$$\text{Exp 4} = 20,625 \quad \text{Exp 5} = 20,5$$

$$k_r = 20,6 \text{ L} \cdot \text{mol}^{-2} \cdot \text{s}^{-2}$$

3.3 The reaction rate depends mainly on the slow step:

$$v = k_2 \cdot c(\text{NOCl}_2) \cdot c(\text{NO})$$

Assuming steady state conditions for NOCl_2 :

$$\frac{dc(\text{NOCl}_2)}{dt} = 0 = k_1 \cdot c(\text{NO}) \cdot c(\text{Cl}_2) - k_{-1} \cdot c(\text{NOCl}_2) - k_2 \cdot c(\text{NOCl}_2) \cdot c(\text{NO})$$

$$\frac{k_1 \cdot c(\text{NO}) \cdot c(\text{Cl}_2)}{k_{-1} + k_2 \cdot c(\text{NO})} = c(\text{NOCl}_2)$$

$$v = k_2 \cdot c(\text{NOCl}_2) \cdot c(\text{NO}) = \frac{k_2 \cdot k_1 \cdot c(\text{NO}) \cdot c(\text{Cl}_2) \cdot c(\text{NO})}{k_{-1} + k_2 \cdot c(\text{NO})}$$

$$v = \frac{k_2 \cdot k_1 \cdot c^2(\text{NO}) \cdot c(\text{Cl}_2)}{k_{-1} + k_2 \cdot c(\text{NO})}$$

$$\mathbf{3.4} \quad v = k_r \cdot c^2(\text{NO}) \cdot c(\text{Cl}_2) \text{ when } k_{-1} \gg k_2 \cdot c(\text{NO}).$$

This occurs when the nitrogen monoxide concentration is small.

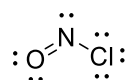
3.5

$$\ln\left(\frac{k'}{k}\right) = -\frac{E_A}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

$$\ln\left(\frac{k'}{20,6}\right) = -\frac{22000}{8,314}\left(\frac{1}{310} - \frac{1}{300}\right)$$

$$k' = 27,4 \text{ L} \cdot \text{mol}^{-2} \cdot \text{s}^{-2}$$

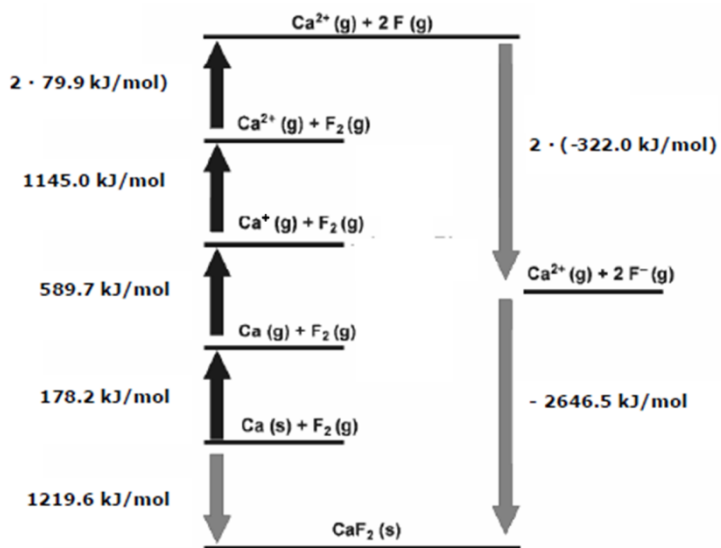
3.6



polar molecule

Problem 4

4.1



Lattice energy of calcium fluoride: -2646,5 kJ/mol

4.2

$$c(\text{Ca}^{2+}) \cdot c(\text{F}^-)^2 = K_{\text{ps}}$$

$$c(\text{Ca}^{2+}) = \frac{1}{2}c(\text{F}^-)$$

$$\frac{1}{2}c(\text{F}^-) \cdot c(\text{F}^-)^2 = K_{\text{ps}} = 10^{-10,52} = 3 \cdot 10^{-11}$$

$$c(\text{F}^-) = \sqrt[3]{6 \cdot 10^{-11}} = 3,92 \cdot 10^{-4} \text{ mol/L}$$

$$S(\text{CaF}_2) = \frac{1}{2}c(\text{F}^-) \cdot M(\text{CaF}_2) = \frac{1}{2} \cdot 3,92 \cdot 10^{-4} \text{ mol/L} \cdot 78,08 \text{ g/mol}$$

$$S(\text{CaF}_2) = 1,53 \cdot 10^{-4} \text{ mg/L}$$

Problem 5

5.1. Which acid must be stored in plastic flasks instead of glass flasks?

A) ☒ Hydrofluoric acid

B) ☐ Nitric acid

C) ☐ Phosphoric acid

D) ☐ Sulfuric acid

5.2. In an experiment to prepare bromobenzene according to: $\text{C}_6\text{H}_6 + \text{Br}_2 = \text{C}_6\text{H}_5\text{Br} + \text{HBr}$, a student departed from 20,0 g of C_6H_6 with 0,310 mol de bromo. If 28,0 g of $\text{C}_6\text{H}_5\text{Br}$ were obtained, ¿which is the yield percentage?

A) ☐ 31,5%

B) ☐ 40,3%

C) ☐ 57,6%

D) ☒ 69,7%

5.3. A 225 mL sample of H_2 is collected over water at 25°C and 735 mmHg. Which expression allow to calculate the dry volume of H_2 at 0°C and 1 atmosphere? The vapor pressure of water at 25°C is 24 mmHg.

A) ☐ $V = \frac{225 \times 760 \times 298}{(735 - 24) \times 273}$

B) ☒ $V = \frac{225 \times (735 - 24) \times 273}{760 \times 298}$

C) ☐ $V = \frac{225 \times 273 \times 760}{(735 + 24) \times 298}$

D) ☐ $V = \frac{225 \times (735 + 24) \times 298}{760 \times 273}$

5.4. Bromine boils at 59°C with $\Delta H_{\text{vap}} = 29,6 \text{ kJ} \cdot \text{mol}^{-1}$, which is the value for ΔS_{vap} ?

A) ☐ $11,2 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$

B) ☒ $89,2 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$

C) ☐ $501 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$

D) ☐ $1,75 \text{ kJ} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$

5.5. How many unpaired electrons are there in a Co^{3+} gas ion at the ground state?

A) ☐ 1

B) ☐ 3

C) ☒ 4

D) ☐ 5

5.6. Which solution produces a black precipitate when adding an aqueous solution of copper(II)?

A) ☐ NH_3

B) ☒ $(\text{NH}_4)_2\text{S}$

C) ☐ K_2SO_4

D) ☐ NaOH

5.7. Which oxide is the best reducing reagent?

A) ☐ CO_2

B) ☐ NO_2

C) ☐ SiO_2

D) ☒ SO_2

5.8. Methylamine, CH_3NH_2 , reacts with oxygen to form CO_2 , N_2 and H_2O . Which amount of oxygen (in moles) is needed to react completely with 1,00 mol of methylamine?

A) ☒ 2,25

B) ☐ 2,50

C) ☐ 3,00

D) ☐ 4,50

5.9. Toluene, C_7H_8 , is added to gas to increase the octane-grade. Which is the ration air volume to toluene(g) for the complete combustion of toluene? (Assume that the air is composed of 20% O_2)

A) 9/1

B) 11/1

C) 28/1

D) X 45/1

5.10. Which of the following elements has the highest second ionization energy?

A) X Na

B) Mg

C) Al

D) K

Problem 6

6.1 Seger's formula for a pigment containing 85% of frite and 15% of kaolin.

In 100 g of pigment the are 85 g of frite and 15 g of kaolin:

Frite:

$85 \cdot 0,6 = 51$ g of feldspar; $51/556,67 = 9,162 \cdot 10^{-2}$ moles of feldspar

$9,162 \cdot 10^{-2}$ moles de K_2O y $9,162 \cdot 10^{-2}$ moles de Al_2O_3 .

$6 \cdot 9,162 \cdot 10^{-2} = 0,5497$ moles of SiO_2

$85 \cdot 0,2 = 17$ g of borax; $17/381,67 = 4,458 \cdot 10^{-2}$ moles of borax

$2 \cdot 4,458 \cdot 10^{-2} = 8,915 \cdot 10^{-2}$ moles of B_2O_3

$4,458 \cdot 10^{-2}$ moles of Na_2O

$85 \cdot 0,1 = 8,5$ g of creta; $8,5/100,088 = 8,493 \cdot 10^{-2}$ moles of creta

$8,493 \cdot 10^{-2}$ moles of CaO

$85 \cdot 0,07 = 5,95$ g of quartz; $5,95/60,084 = 9,90 \cdot 10^{-2}$ moles of quartz

$9,90 \cdot 10^{-2}$ moles of SiO_2

$85 \cdot 0,03 = 2,55$ g of boric acid; $2,55/61,832 = 4,12 \cdot 10^{-2}$ moles of boric acid

$1/2 \cdot 4,12 \cdot 10^{-2} = 2,062 \cdot 10^{-2}$ moles of B_2O_3

Kaolin:

15 g de kaolin; $15/258,158 = 5,81 \cdot 10^{-2}$ moles of kaolin

$5,81 \cdot 10^{-2}$ moles de Al_2O_3 y $2 \cdot 5,81 \cdot 10^{-2} = 0,11621$ moles de SiO_2

Total moles of oxides:

Moles of basic oxides:

$9,162 \cdot 10^{-2}$ moles of K_2O + $8,493 \cdot 10^{-2}$ moles of CaO + $4,4 \cdot 10^{-2}$ moles of Na_2O = 0,2211 moles (M_2O, MO)

Moles of amphoteric oxides:

$$9,162 \cdot 10^{-2} \text{ moles of Al}_2\text{O}_3 + 5,81 \cdot 10^{-2} \text{ moles of Al}_2\text{O}_3 = 0,1497 \text{ moles (M}_2\text{O}_3)$$

Moles of acid oxides:

$$0,5497 + 0,099 + 0,1162 = 0,7649 \text{ moles of SiO}_2$$

$$8,915 \cdot 10^{-2} + 2,062 \cdot 10^{-2} = 0,1098 \text{ of B}_2\text{O}_3$$

$$0,7649 + 0,1098 = 0,8747 \text{ moles (MO}_2, \text{B}_2\text{O}_3)$$

Seeger's formula:

$$(\text{M}_2\text{O}, \text{MO}) \cdot 0,6771 (\text{M}_2\text{O}_3) \cdot 3,956 (\text{MO}_2, \text{B}_2\text{O}_3)$$

6.2 a)

$$10 \text{ g of Y}_{0,95}\text{Nd}_{0,05}\text{Al}_{0,95}\text{Cr}_{0,05}\text{O}_3 ; 10/167,901 = 5,956 \cdot 10^{-2} \text{ moles}$$

$$(0,95/2) \cdot 5,956 \cdot 10^{-2} = 2,829 \cdot 10^{-2} \text{ moles Y}_2\text{O}_3$$

$$2,829 \cdot 10^{-2} \text{ moles Y}_2\text{O}_3 \cdot 225,807 \text{ (g/mol)} \cdot (100/99,9) = 6,3946 \text{ g Y}_2\text{O}_3 \text{ impure}$$

$$(0,05/2) \cdot 5,956 \cdot 10^{-2} \cdot 336,477 \cdot (100/99,8) = 0,5023946 \text{ g Nd}_2\text{O}_3 \text{ impure}$$

$$(0,95/2) \cdot 5,956 \cdot 10^{-2} \cdot 101,96 \cdot (100/99,5) = 2,899 \text{ g Al}_2\text{O}_3 \text{ impure}$$

$$(0,05/2) \cdot 5,956 \cdot 10^{-2} \cdot 151,989 \cdot (100/99) = 0,2286 \text{ g Cr}_2\text{O}_3 \text{ impure}$$

b)

$$10 \text{ g de CoAl}_{1,25}\text{Cr}_{0,75}\text{O}_4; 10/195,653 = 5,111 \cdot 10^{-2} \text{ moles}$$

$$(1/3) \cdot 5,111 \cdot 10^{-2} = 1,704 \cdot 10^{-2} \text{ moles Co}_3\text{O}_4$$

$$1,704 \cdot 10^{-2} \text{ moles Cr}_3\text{O}_4 \cdot 240,795 \text{ (g/mol)} \cdot (100/98) = 4,1861 \text{ g Co}_3\text{O}_4 \text{ impure}$$

$$(1,25/2) \cdot 5,111 \cdot 10^{-2} \cdot 101,96 \cdot (100/99,5) = 3,2734 \text{ g Al}_2\text{O}_3 \text{ impure}$$

$$(0,75/2) \cdot 5,111 \cdot 10^{-2} \cdot 151,989 \cdot (100/99) = 2,9425 \text{ g Cr}_2\text{O}_3 \text{ impure}$$

Problem 7

7.1 Lattice energy of NaH

$$\Delta H_f^\circ(\text{NaH}) = \Delta H_{\text{sub}}^\circ(\text{Na}) + 0,5 \cdot \Delta H_{\text{dis}}^\circ(\text{H}_2) + I(\text{Na}) + A(\text{H}) + U(\text{NaH})$$

$$-56 = 107 + 0,5 \cdot 436 + 496 - 73 + U(\text{NaH})$$

$$U(\text{NaH}) = -804 \text{ kJ} \cdot \text{mol}^{-1}$$

7.2

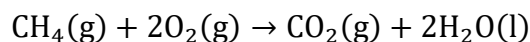
$$\Delta H_r^\circ = \sum \Delta H^\circ(\text{broken bonds}) - \sum \Delta H^\circ(\text{formed bonds})$$

$$\Delta H_r^\circ = 4 \cdot \Delta H^\circ(\text{O} - \text{H}) + 2 \cdot \Delta H^\circ(\text{O} - \text{O}) - 4 \cdot \Delta H^\circ(\text{O} - \text{H}) - \Delta H^\circ(\text{O} = \text{O})$$

$$\Delta H_r^\circ = 2 \cdot 146 - 497$$

$$\Delta H_r^\circ = -206 \text{ kJ} \cdot \text{mol}^{-1}$$

7.3



$$\Delta H_r^\circ = \sum v \cdot \Delta H_f^\circ$$

$$\Delta H_c^\circ(\text{CH}_4) = \Delta H_f^\circ(\text{CO}_2) + 2 \cdot \Delta H_f^\circ(\text{H}_2\text{O}) - \Delta H_f^\circ(\text{CH}_4) - 2 \cdot \Delta H_f^\circ(\text{O}_2)$$

$$\Delta H_c^\circ(\text{CH}_4) = -394 + 2 \cdot (-286) - (-75)$$

$$\Delta H_c^\circ(\text{CH}_4) = -891 \text{ kJ} \cdot \text{mol}^{-1}$$

7.4

$$M(\text{hidrato}) = 8 \cdot M(\text{CH}_4) + 46 \cdot M(\text{H}_2\text{O})$$

$$M(\text{hidrato}) = 8 \cdot 16,0 + 46 \cdot 18,0 = 956 \text{ g} \cdot \text{mol}^{-1}$$

$$n(\text{hidrato}) = \frac{m(\text{hydrate})}{M(\text{hydrate})} = \frac{1,00}{956} = 0,00105 \text{ mol}$$

$$n(\text{CH}_4) = 8 \cdot n(\text{hydrate}) = 8 \cdot 0,00105 = 0,00837 \text{ mol}$$

$$\Delta H_r^\circ = n(\text{CH}_4) \cdot \Delta H_c^\circ(\text{CH}_4) = 0,00837 \cdot (-891) = -7,46 \text{ kJ}$$

7.5

$$n(\text{CH}_4) = \frac{m(\text{CH}_4)}{M(\text{CH}_4)} = \frac{5 \cdot 10^{11} \cdot 10^6}{16,0} = 3,13 \cdot 10^{16} \text{ mol}$$

$$\Delta H_r^\circ = n(\text{CH}_4) \cdot \Delta H_c^\circ(\text{CH}_4)$$

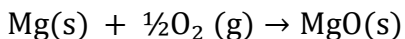
$$\Delta H_r^\circ = 3,13 \cdot 10^{16} \cdot (-891) = -2,78 \cdot 10^{19} \text{ kJ} = -2,78 \cdot 10^{22} \text{ J}$$

$$Q_{\text{atm}} = -\Delta H_r^\circ = 2,78 \cdot 10^{22} \text{ J}$$

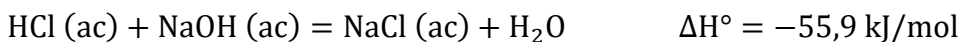
$$\Delta T = \frac{Q_{\text{atm}}}{C} = \frac{2,78 \cdot 10^{22}}{4,21 \cdot 10^{21}} = 6,96 \text{ K} \approx 7 \text{ K}$$

Problem 8

8.1



8.2 The first step is to determine the limiting substance:



$$n(\text{HCl}) = 0,0496 \text{ L} * 1,01 \text{ mol/L} = 0,0501 \text{ mol}$$

$$n(\text{NaOH}) = 0,0501 \text{ L} * 0,998 \text{ mol/L} = 0,0500 \text{ mol}$$

As the stoichiometry is 1:1, NaOH is the limiting substance and 0,0500 moles of water are formed.

Then, the heat released in the neutralization reaction is

$$\Delta H = -55,9 \text{ kJ/mol} * 0,0500 \text{ mol} = -2,795 \text{ kJ}$$

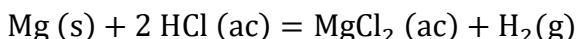
The total volume of the solution is 49,6 mL + 50,1 mL = 99,7 mL. Because the density is 1,0 g/mL, the mass of the solution is 99,7 g. Using the value of the specific heat capacity of the solution, the heat absorbed is:

$$Q_{\text{dis}} = 99,7 \text{ g} * 4,025 \text{ J} \cdot \text{g}^{-1} \cdot ^\circ\text{C}^{-1} * 6,40^\circ\text{C} = 2568 \text{ J}$$

Therefore, from the total heat produced in the neutralization reaction, the calorimeter absorbs $Q_{\text{cal}} = \Delta H - Q_{\text{dis}} = 2795 \text{ J} - 2568 \text{ J} = 227 \text{ J}$. Then, the caloric heat of the calorimeter is:

$$C_{\text{cal}} = \frac{227 \text{ J}}{6,40^\circ\text{C}} = 35,5 \text{ J} \cdot ^\circ\text{C}^{-1}$$

8.3 The reaction of magnesium with acid is



$$n(\text{Mg}/2) = 2 * \frac{0,221 \text{ g}}{24,31 \text{ g/mol}} = 0,0182 \text{ mol}$$

$$n(\text{HCl}/1) = 0,0499 \text{ L} * 1,01 \text{ mol/L} = 0,0504 \text{ mol}$$

The limiting substance is magnesium and the amount of substance that reacts is 0,00909 mol.

The total mass is 99,6 g solution + 0,221 g Mg = 99,821 g

The total heat is the heat absorbed by the solution + the heat absorbed by the calorimeter:

$$Q_{\text{dis}} = 99,821 \text{ g} * 3,862 \text{ J} \cdot \text{g}^{-1} \cdot ^\circ\text{C}^{-1} * 9,67^\circ\text{C} = 3728 \text{ J}$$

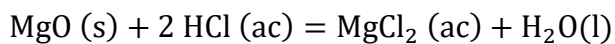
$$Q_{\text{cal}} = 35,5 \text{ J} \cdot ^\circ\text{C}^{-1} * 9,67^\circ\text{C} = 343 \text{ J}$$

$$\Delta H = Q_{\text{cal}} + Q_{\text{dis}} = 3728 \text{ J} + 343 \text{ J} = 4071 \text{ J}$$

As it is an exothermic process, the value of the standard energy for the reaction is:

$$\Delta H^\circ = \frac{\Delta H}{n(\text{Mg})} = \frac{-4071 \text{ J}}{0,00909 \text{ mol}} = -447,9 \text{ kJ} \cdot \text{mol}^{-1}$$

8.3 The procedure here is analogous to the previous task. The reaction of magnesium oxide with acid is:



$$n(\text{MgO}/2) = 2 * \frac{0,576 \text{ g}}{40,31 \text{ g/mol}} = 0,0286 \text{ mol}$$

$$n(\text{HCl}/1) = 0,0499 \text{ L} * 1,01 \text{ mol/L} = 0,0515 \text{ mol}$$

The limiting substance is magnesium oxide and the amount of substance that reacts is 0,0143 mol.

The total mass is 101,1 g solution + 0,576 g Mg = 101,676 g

The total heat is the heat absorbed by the solution + heat absorbed by the calorimeter:

$$Q_{\text{dis}} = 101,676 \text{ g} * 3,862 \text{ J} \cdot \text{g}^{-1} \cdot ^\circ\text{C}^{-1} * 4,72^\circ\text{C} = 1853 \text{ J}$$

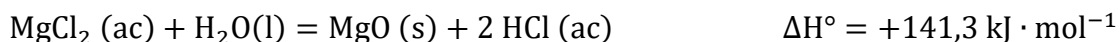
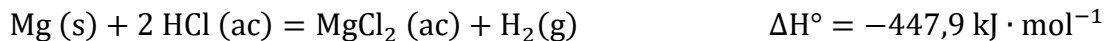
$$Q_{\text{cal}} = 35,5 \text{ J} \cdot ^\circ\text{C}^{-1} * 4,72^\circ\text{C} = 168 \text{ J}$$

$$\Delta H = Q_{\text{cal}} + Q_{\text{dis}} = 1853 \text{ J} + 168 \text{ J} = 2021 \text{ J}$$

As it is an exothermic process, the value of the standard enthalpy change for the reaction is:

$$\Delta H^\circ = \frac{-2021 \text{ J}}{0,0143 \text{ mol}} = -141,3 \text{ kJ} \cdot \text{mol}^{-1}$$

8.5 Ordering the previous equations conveniently, in addition to the formation of water, one gets:



The standard enthalpy for the formation of magnesium oxide is -592,4 kJ·mol⁻¹.

Problem 9**9.1**

$$M(\text{HCl}) = A_r(\text{H}) + A_r(\text{Cl}) = 1,01 + 35,45 = 36,46 \text{ g} \cdot \text{mol}^{-1}$$

$$c(\text{HCl}) = \frac{\omega_{\%} \cdot \rho_D \cdot 10}{M(\text{HCl})} = \frac{37 \cdot 1,19 \cdot 10}{36,46} = 12,1 \text{ mol} \cdot \text{L}^{-1}$$

$$V(\text{HCl}) = \frac{c'(\text{HCl}) \cdot V'(\text{HCl})}{c(\text{HCl})} = \frac{0,50 \cdot 500}{12,1} = 20,7 \text{ mL}$$

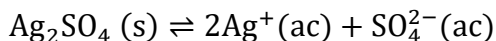
9.2

$$\alpha_2 = \frac{K_1 K_2}{[\text{H}^+]^2 + K_1 [\text{H}^+] + K_1 K_2} = 1,3 \cdot 10^{-14}$$

$$[\text{S}^{2-}] = \alpha_2 \cdot c(\text{S}^{2-}) = 1,3 \cdot 10^{-14} \cdot 0,1 = 1,3 \cdot 10^{-15} \text{ mol} \cdot \text{L}^{-1}$$

$$[\text{Pb}^{2+}][\text{S}^{2-}] = 5 \cdot 10^{-5} \cdot 1,3 \cdot 10^{-15} = 6,5 \cdot 10^{-20}$$

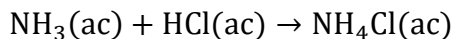
As this value is higher than $K_{ps}(\text{PbS})$, the precipitate is formed.

9.3

$$[\text{Ag}^+] = 2S \quad [\text{SO}_4^{2-}] = S$$

$$K_{ps} = [\text{Ag}^+][\text{SO}_4^{2-}] = (2S)^2 \cdot S = 4S^3$$

$$S = \sqrt[3]{\frac{K_{ps}}{4}} = \sqrt[3]{\frac{1,2 \cdot 10^{-5}}{4}} = 0,014 \text{ mol} \cdot \text{L}^{-1}$$

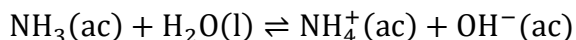
9.4

$$n(\text{HCl}) = c(\text{HCl}) \cdot V(\text{HCl}) = 0,6 \cdot 10 = 6,0 \text{ mmol}$$

$$n(\text{NH}_3) = c(\text{NH}_3) \cdot V(\text{NH}_3) = 0,5 \cdot 20 = 10,0 \text{ mmol}$$

$$n(\text{NH}_4\text{Cl}) = n(\text{HCl}) = 6,0 \text{ mmol}$$

$$n(\text{NH}_3)_{\text{exc}} = 10,0 - 6,0 = 4,0 \text{ mmol}$$



$$[\text{OH}^-] = K_b \cdot \frac{[\text{NH}_3]}{[\text{NH}_4^+]} = 1,8 \cdot 10^{-5} \cdot \frac{4,0/30}{6,0/30} = 1,2 \cdot 10^{-5} \text{ mol} \cdot \text{L}^{-1}$$

$$\text{pH} = 14 - \text{pOH} = 14 - \log(1,2 \cdot 10^{-5}) \approx 9,1$$

Problem 10

10.1

$$b(\text{A}/\text{H}_2\text{O}) = \frac{n(\text{A})}{m(\text{H}_2\text{O})} = \frac{m(\text{A})}{M(\text{A}) \cdot m(\text{H}_2\text{O})} = \frac{30 \text{ g}}{180 \text{ g/mol} \cdot 0,1 \text{ kg}} = 1,667 \text{ mol/kg}$$

$$\Delta T_f = K_{cr} \cdot b(\text{A}/\text{H}_2\text{O}) = 1,86 \text{ K} \cdot \text{kg/mol} \cdot 1,667 \text{ mol/kg} = 3,1 \text{ K}$$

$$T_f = 0^\circ\text{C} - \Delta T_f = -3,1^\circ\text{C}$$

10.2

$$b(\text{A}/\text{H}_2\text{O}) = \Delta T_f \div K_{cr} = 7 \text{ K} \div 1,86 \text{ K} \cdot \text{kg/mol} = 3,763 \text{ mol/kg}$$

$$m(\text{A}) = b(\text{A}/\text{H}_2\text{O}) \cdot M(\text{A}) \cdot m(\text{H}_2\text{O}) = 3,763 \text{ mol/kg} \cdot 180 \text{ g/mol} \cdot 5 \text{ kg} = 3387 \text{ g}$$

10.3

$$b(\text{C}/\text{H}_2\text{O}) = \Delta T_f \div K_{cr} = 3,03 \text{ K} \div 1,86 \text{ K} \cdot \text{kg/mol} = 1,629 \text{ mol/kg}$$

$$M(\text{C}) = \frac{m(\text{C})}{b(\text{C}/\text{H}_2\text{O}) \cdot m(\text{H}_2\text{O})} = \frac{300 \text{ g}}{1,629 \text{ mol/kg} \cdot 0,7 \text{ kg}} = 263 \text{ g/mol}$$

10.4 Calculation of the molar mass of the mixture gives as result the same value of the previous task, 263 g/mol, because the solution has equal mass and equal freezing point. Then, in the mixture:

$$xM(\text{A}) + (1 - x)M(\text{B}) = 263 \text{ g/mol}$$

$$180x + 342(1 - x) = 263$$

$$x = 0,4877$$

As x is the molar fraction of A in the mixture, the mass fraction is:

$$\omega(\text{A}) = \frac{180x}{263} = 0,334$$

$$m(\text{A}) = \omega(\text{A}) \cdot m(\text{mezcla}) = 0,334 \cdot 300 \text{ g} = 100,2 \text{ g}$$

$$b(\mathbf{A}/\text{H}_2\text{O}) = \frac{m(\mathbf{A})}{M(\mathbf{A}) * m(\text{H}_2\text{O})} = \frac{100,2 \text{ g}}{180 \text{ g/mol} * 0,7 \text{ kg}} = 0,795 \text{ mol/kg}$$

10.5

$$b(\text{mixture}/\text{H}_2\text{O}) = \Delta T_f \div K_{cr} = 2,5 \text{ K} \div 1,86 \text{ K} \cdot \text{kg/mol} = 1,344 \text{ mol/kg}$$

$$n(\text{mixture}) = b(\text{mixture}/\text{H}_2\text{O}) * m(\text{H}_2\text{O}) = 1,344 \text{ mol/kg} * 0,4 \text{ kg} = 0,538 \text{ mol}$$

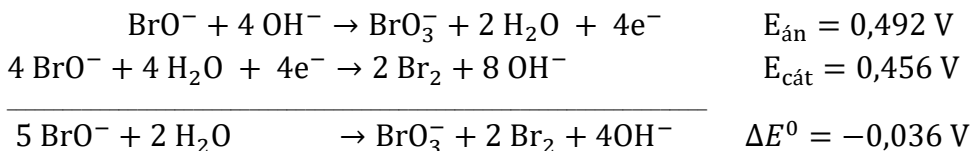
Because masses of **A** and **B** are the same and sum of the amount of substance gives 0,538 mol:

$$\frac{m(\mathbf{A})}{M(\mathbf{A})} + \frac{m(\mathbf{B})}{M(\mathbf{B})} = \frac{m(\mathbf{A})}{180} + \frac{m(\mathbf{A})}{342} = 0,538$$

$$m(\mathbf{A}) = 63,4 \text{ g}$$

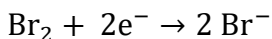
Problem 11

11.1 Thermodynamic stability of the species BrO^- with respect to the dismutation to the species BrO_3^- and Br_2 .



$\Delta E^0 < 0$; as $\Delta G^0 = -nF\Delta E^0$, then $\Delta G^0 > 0$: the process is not spontaneous. Species BrO^- is stable.

11.2 As in the equation corresponding to the redox pair



there are no protons or hydroxyl, the standard redox potential is not affected by the pH, and in consequence, in acid media $E^0 \text{Br}_2/\text{Br}^- = 1,065 \text{ V}$.

11.3 Reduction potential of bromate ion to bromide

$$\Delta G^0(\text{BrO}_3^-/\text{Br}^-) = \Delta G^0(\text{BrO}_3^-/\text{BrO}^-) + \Delta G^0(\text{BrO}^-/\text{Br}_2) + \Delta G^0(\text{Br}_2/\text{Br}^-)$$

$$\Delta E^0(\text{BrO}_3^-/\text{Br}^-) = \frac{4\Delta E^0(\text{BrO}_3^-/\text{BrO}^-) + \Delta E^0(\text{BrO}^-/\text{Br}_2) + \Delta E^0(\text{Br}_2/\text{Br}^-)}{6}$$

$$\Delta E^0(\text{BrO}_3^-/\text{Br}^-) = 0,581 \text{ V}$$

Problem 12

12.1 Empirical formula of compounds

A: C_xH_yO_z

$$x:y:z = \frac{41,38}{12} : \frac{3,45}{1} : \frac{55,17}{16} \approx 3,45:3,45:3,45 = 1:1:1$$

B: C_mH_nO_p

$$m:n:p = \frac{55,81}{12} : \frac{6,97}{1} : \frac{37,22}{16} \approx 4,65:6,97:2,33 \approx 2:3:1$$

D: C_aH_bO_c

$$a:b:c = \frac{35,82}{12} : \frac{4,48}{1} : \frac{59,70}{16} \approx 2,99:4,48:3,73 \approx 4:6:5$$

Then the final empirical formula are: **A** (CHO), **B** (C₂H₃O) y **D** (C₄H₆O₅)

$$n(\mathbf{D}/_{z^*}) = n(\text{KOH}/_1) * V(\text{KOH}) = 2 \text{ mol/L} * 0,02 \text{ L} = 0,04 \text{ mol}$$

$$M(\mathbf{D}/_{z^*}) = m(\mathbf{D}) \div n(\mathbf{D}/_{z^*}) = 2,68 \text{ g} \div 0,04 \text{ mol} = 67 \text{ g/mol}$$

The equivalent number z^* indicates the number of acidic protons of compound **D**. For $z^* = 1, 2, 3 \dots$ $M(\mathbf{D}) = 67, 134, 201 \dots \text{g/mol}$, respectively. Taking as reference the empirical formula C₄H₆O₅, which corresponds to $M(\mathbf{D}) = 134 \text{ g/mol}$, one can conclude that **D** is a diprotic acid of molecular formula C₄H₆O₅.

As **D** is obtained after adding bromhydric acid to **A** (forming **C**) and after carrying out the hydrolysis, both **A** and **C** must have the same number of carbon atoms than **D**, and **A** has at least one multiple bonds. The molecular formula of **A** should be C₄H₄O₄, which corresponds to maleic acid with two carboxylic groups in *cis* position, therefore it can lose water to form the correspondent anhydride.

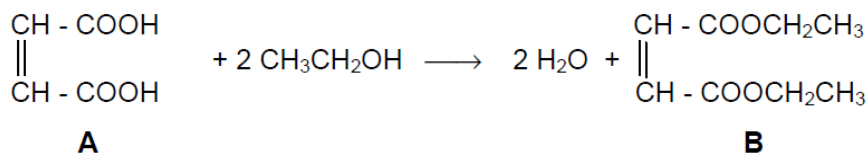
A: C₄H₄O₄

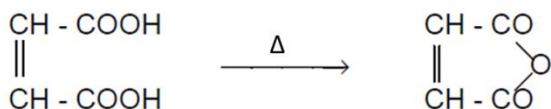
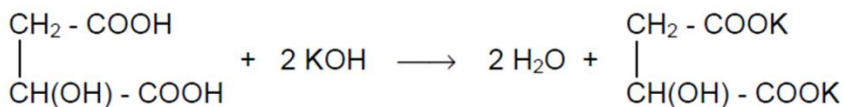
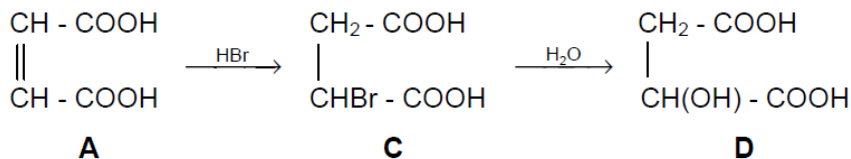
B: C₈H₁₂O₄

C: C₄H₅O₄Br

D: C₄H₆O₅

12.2 Equations:





Problem 13

13.1 Solubility condition: $[\text{Tl}^+]^2 \cdot [\text{S}^{2-}] \leq K_{\text{ps}}(\text{Tl}_2\text{S}) = 1 \cdot 10^{-20}$

$$[\text{Tl}^+] = c(\text{Tl}^+) = 2 * n(\text{Tl}_2\text{S}) \div V = 0,2 \text{ mol/L}$$

$$c(\text{S}^{2-}) = [\text{S}^{2-}] + [\text{HS}^-] + [\text{H}_2\text{S}] = [\text{S}^{2-}] \left(1 + \frac{[\text{H}^+]}{K_2} + \frac{[\text{H}^+]^2}{K_1 K_2} \right) = 0,1 \text{ mol/L}$$

$$[\text{S}^{2-}] = \frac{0,1}{1 + \frac{[\text{H}^+]}{K_2} + \frac{[\text{H}^+]^2}{K_1 K_2}} = \frac{0,1}{1 + \frac{1}{1 \cdot 10^{-13}} + \frac{1^2}{1 \cdot 10^{-13} * 1 \cdot 10^{-7}}} \approx 1 \cdot 10^{-21}$$

$$[\text{Tl}^+]^2 [\text{S}^{2-}] = 4 \cdot 10^{-23} < K_{\text{ps}}(\text{Tl}_2\text{S})$$

Then, Tl_2S will be soluble in 1 L of non-coordinated monoprotic strong acid of 1 M concentration.

13.2

- For CuS in HCl (non-complexing, non-oxidant):

$$[\text{Cu}^{2+}] = c(\text{Cu}^{2+}) = 0,1 \text{ mol/L}$$

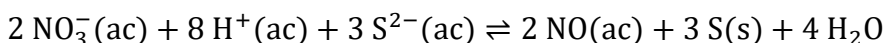
In analogy to the previous task, one gets to:

$$[\text{S}^{2-}] = \frac{0,1}{1 + \frac{[\text{H}^+]}{K_2} + \frac{[\text{H}^+]^2}{K_1 K_2}} = \frac{0,1}{1 + \frac{1}{1 \cdot 10^{-13}} + \frac{1^2}{1 \cdot 10^{-13} * 1 \cdot 10^{-7}}} \approx 1 \cdot 10^{-21}$$

$$[\text{Cu}^{2+}][\text{S}^{2-}] = 1 \cdot 10^{-23} > K_{\text{ps}}(\text{CuS}) = 1 \cdot 10^{-35}$$

Then, CuS is not soluble in one liter of HCl 1 M.

- For CuS in HNO₃ (oxidant acid), oxidation of S²⁻ to S (s) happens:



The equilibrium constant for this reaction can be found according to:

$$\log K = -\frac{\Delta G}{RT} = \frac{nF\Delta E}{RT} = \frac{n(E_{\text{NO}_3^-/\text{NO}} - E_{\text{S/S}^{2-}})}{0,0591} = \frac{6(0,96 + 0,48)}{0,0591} = 146$$

This equilibrium constant can be represented in the following form:

$$K = \frac{[\text{NO}]^2[\text{S}]}{[\text{NO}_3^-]^2[\text{H}^+]^8[\text{S}^{2-}]^3}$$

Where [S] represents the amount of substance of S that forms divided for the solution volume (even if solid, this consideration is necessary for later calculations)

Isolating [S], in the redox equilibrium:

$$[\text{S}] = \sqrt[3]{\frac{K[\text{NO}_3^-]^2[\text{H}^+]^8[\text{S}^{2-}]^3}{[\text{NO}]^2}}$$

Because [NO₃⁻] = [H⁺] = 1, the previous equation simplifies to:

$$[\text{S}] = [\text{S}^{2-}]^3 \sqrt[3]{\frac{K}{[\text{NO}]^2}}$$

Then, the total concentration of sulfide coming from CuS is:

$$c(\text{S}^{2-}) = [\text{S}^{2-}] + [\text{HS}^-] + [\text{H}_2\text{S}] + [\text{S}] = [\text{S}^{2-}] \left(1 + \frac{[\text{H}^+]}{K_2} + \frac{[\text{H}^+]^2}{K_1 K_2} + \sqrt[3]{\frac{K}{[\text{NO}]^2}} \right)$$

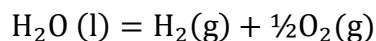
$$[\text{S}^{2-}] = \frac{0,1}{1 + \frac{[\text{H}^+]}{K_2} + \frac{[\text{H}^+]^2}{K_1 K_2} + \sqrt[3]{\frac{K}{[\text{NO}]^2}}} \approx 8,62 \cdot 10^{-51}$$

$$[\text{Cu}^{2+}][\text{S}^{2-}] = 8,62 \cdot 10^{-52} < K_{\text{ps}}(\text{CuS}) = 1 \cdot 10^{-35}$$

Therefore, CuS is soluble in 1,0 L of HNO₃ 1 M.

Problem 14

14.1 The decrease in the mass of the solution is caused by two factors: the electrolytic decomposition of water and the crystallization of $\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$. After the electrolysis a saturated solution remains.



In anodic and cathodic semi-reactions, 2 electrons are exchanged by each molecule of water, therefore the water mass that is decomposed will be:

$$m(\text{H}_2\text{O}) = n(\text{H}_2\text{O}) * M\left(\frac{\text{H}_2\text{O}}{2}\right) = \frac{I * t}{F} * \frac{M(\text{H}_2\text{O})}{2} = \frac{5 \text{ A} * 12 * 3600 \text{ s} * 18 \text{ g/mol}}{96485 \text{ C/mol}}$$

$$m(\text{H}_2\text{O}) = 20,15 \text{ g}$$

The mass lost by crystallization is:

$$m(\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}) = 41,9 - 20,15 = 21,75 \text{ g}$$

$$n(\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}) = \frac{m(\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O})}{m(\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O})} = \frac{21,75 \text{ g}}{236,1 \text{ g/mol}} = 0,092 \text{ mol}$$

Before the electrolysis, there was in the solution:

H_2O : 160 g

$\text{Ca}(\text{NO}_3)_2$: 100 g, that is equivalent to $100/164,1 = 0,609 \text{ mol}$

After the electrolysis in the solution remain:

H_2O : $(160 - 20,15 - 4 \times 18 \times 0,092) \text{ g} = 133,23 \text{ g}$ (taking into account the mass that is lost in the electrolysis and in the hydrate that precipitates)

$\text{Ca}(\text{NO}_3)_2$: $(0,609 - 0,0092) \text{ mol} = 0,527 \text{ mol}$

To produce a saturated solution, and x amount of water is needed plus 0,527 mol of hydrate, which provide the additional water:

$$133,23 = x + 4 * 18 * 0,527$$

$$x = 95,29 \text{ g}$$

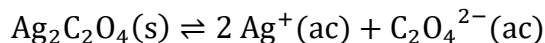
Then, for 100 g of water, the mass of $\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$ needed to reach saturation is:

$$m(\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}) = \frac{100 \text{ g}}{95,29 \text{ g}} * 0,527 \text{ mol} * 236,1 \text{ g/mol} = 130,6 \text{ g}$$

14.2 When dissolving silver oxalate in water, oxalate ions are formed. According to the media pH, oxalate ions will react with H_3O^+ ions. The lowest the pH, the more oxalate ions will react to form

HC_2O_4^- and $\text{H}_2\text{C}_2\text{O}_4$ ions, and the more silver oxalate will dissolve in water. Therefore, the solubility of silver oxalate depends on the pH.

14.3



$$[\text{Ag}^+]^2[\text{C}_2\text{O}_4^{2-}] = K_{\text{ps}}(\text{Ag}_2\text{C}_2\text{O}_4)$$

$$[\text{Ag}^+] = c(\text{Ag}^+) = 2S \text{ and } c(\text{C}_2\text{O}_4^{2-}) = S$$

The total concentration of oxalate species is related to pH according to:

$$c(\text{C}_2\text{O}_4^{2-}) = [\text{C}_2\text{O}_4^{2-}] + [\text{HC}_2\text{O}_4^-] + [\text{H}_2\text{C}_2\text{O}_4] = [\text{C}_2\text{O}_4^{2-}] \left(1 + \frac{[\text{H}^+]}{K_2} + \frac{[\text{H}^+]^2}{K_1 K_2} \right)$$

$$[\text{C}_2\text{O}_4^{2-}] = \frac{c(\text{C}_2\text{O}_4^{2-})}{1 + \frac{[\text{H}^+]}{K_2} + \frac{[\text{H}^+]^2}{K_1 K_2}} = \frac{S}{1 + \frac{[\text{H}^+]}{K_2} + \frac{[\text{H}^+]^2}{K_1 K_2}}$$

Then, the value of $K_{\text{ps}}(\text{Ag}_2\text{C}_2\text{O}_4)$ at pH = 7 is equal to:

$$K_{\text{ps}}(\text{Ag}_2\text{C}_2\text{O}_4) = \frac{S}{1 + \frac{[\text{H}^+]}{K_2} + \frac{[\text{H}^+]^2}{K_1 K_2}} * (2S)^2 = \frac{4S^3}{1 + \frac{[\text{H}^+]}{K_2} + \frac{[\text{H}^+]^2}{K_1 K_2}}$$

$$K_{\text{ps}}(\text{Ag}_2\text{C}_2\text{O}_4) = \frac{4(2,06 \cdot 10^{-4})^3}{1 + \frac{[1 \cdot 10^{-7}]}{6,4 \cdot 10^{-5}} + \frac{[1 \cdot 10^{-7}]^2}{5,9 \cdot 10^{-2} * 6,4 \cdot 10^{-5}}} = 3,49 \cdot 10^{-11}$$

Finally, the solubility at pH = 5

$$S = \sqrt[3]{\frac{K_{\text{ps}}(\text{Ag}_2\text{C}_2\text{O}_4)}{4} \left(1 + \frac{[\text{H}^+]}{K_2} + \frac{[\text{H}^+]^2}{K_1 K_2} \right)}$$

$$S = \sqrt[3]{\frac{3,49 \cdot 10^{-11}}{4} \left(1 + \frac{[1 \cdot 10^{-5}]}{6,4 \cdot 10^{-5}} + \frac{[1 \cdot 10^{-5}]^2}{5,9 \cdot 10^{-2} * 6,4 \cdot 10^{-5}} \right)}$$

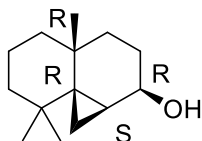
$$S = 2,16 \cdot 10^{-4} \text{ mol/L}$$

Problem 15

15.1.

4 stereoisomers

15.2.



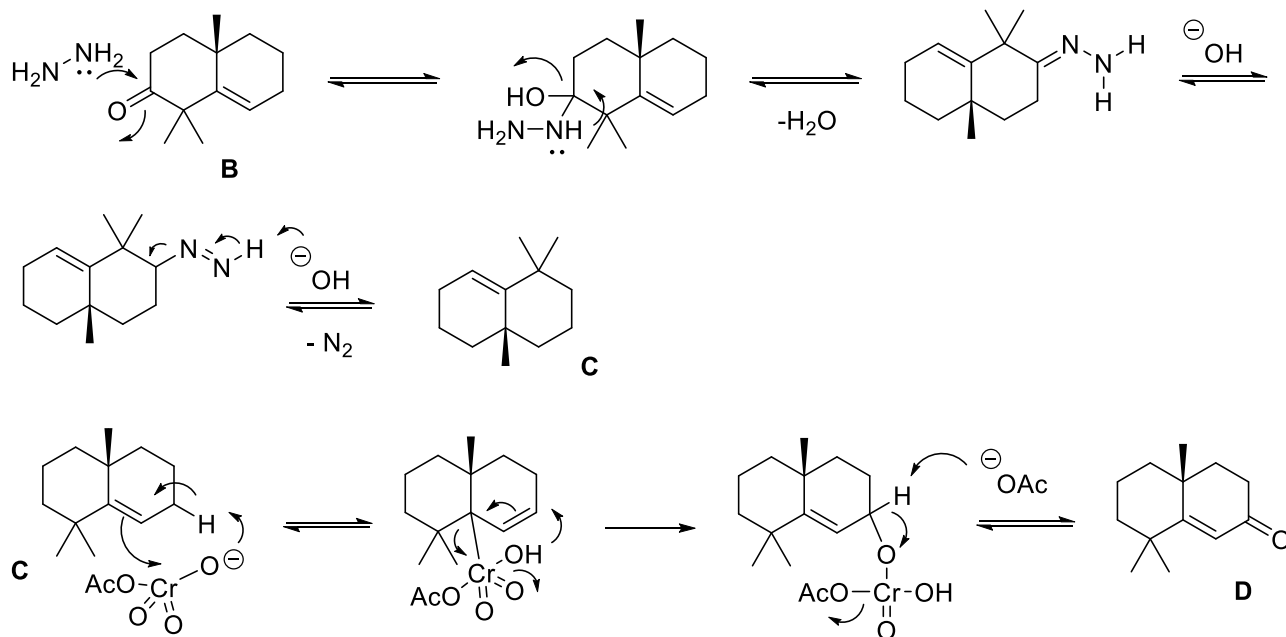
15.3.

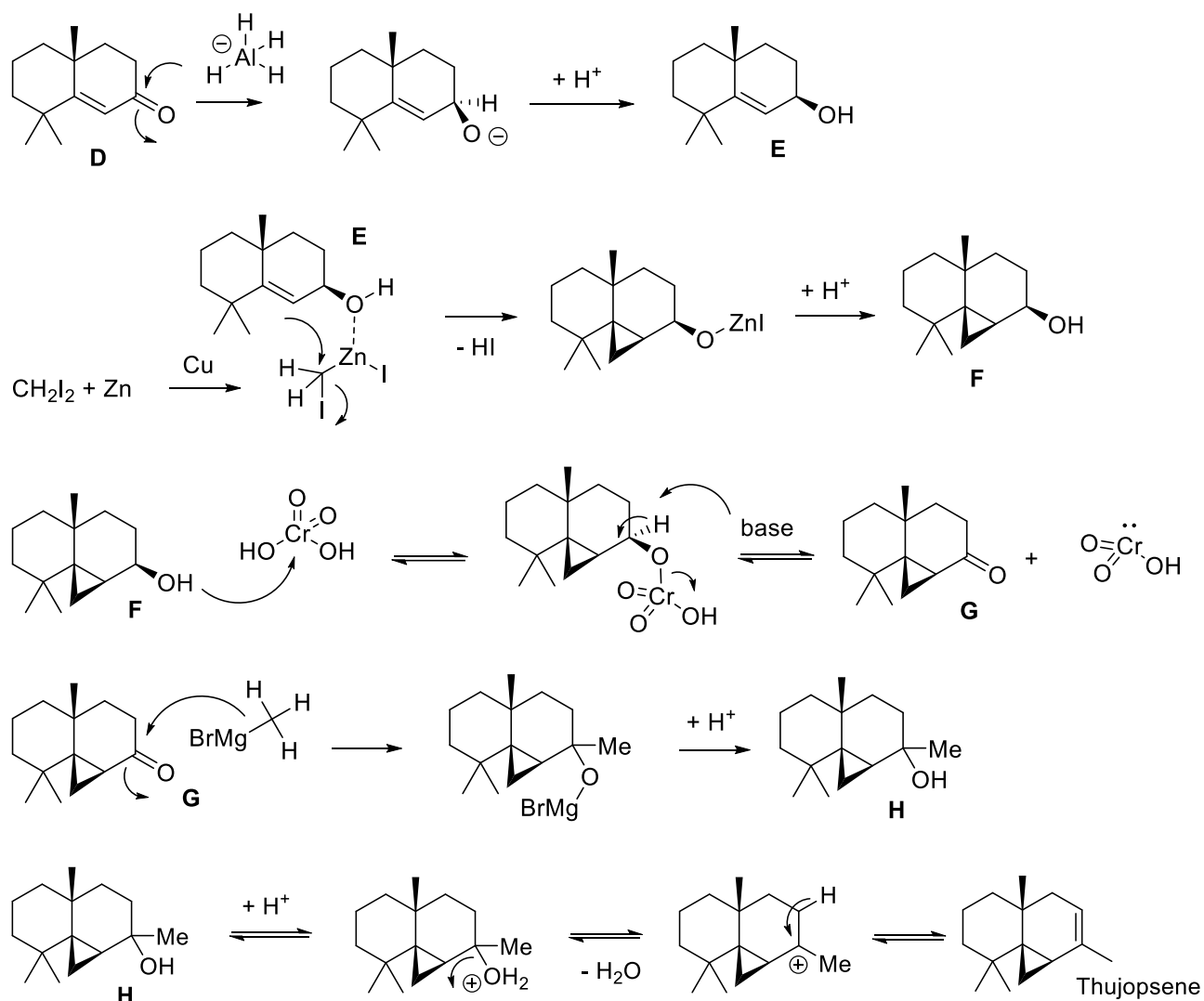
The attack of a hydride to a carbonyl occurs by the least hindered face of cyclic compound **D**, which is opposed to the methyl group.

15.4.

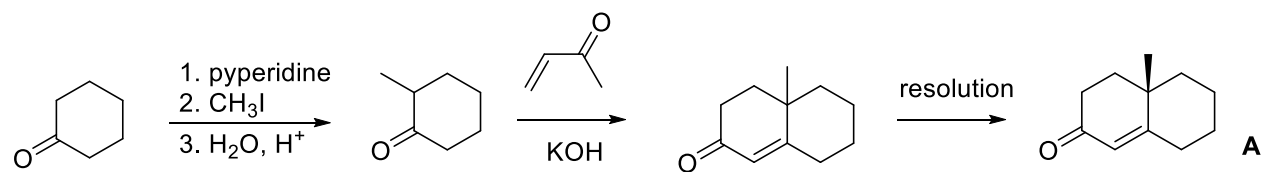
The Simmons-Smith's reaction occurs *syn* with respect to the hydroxyl group. The oxygen atom interacts with a zinc atom of carbenoid IZnCH_2I and stabilizes the transition state.

15.5.



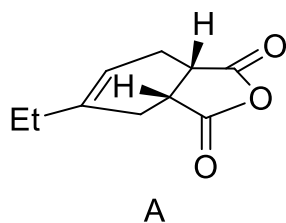


15.6.

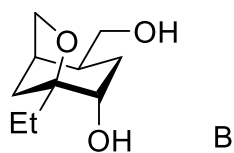


Problem 16

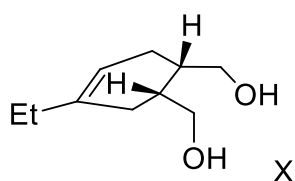
16.1 Diels-Alder's reaction



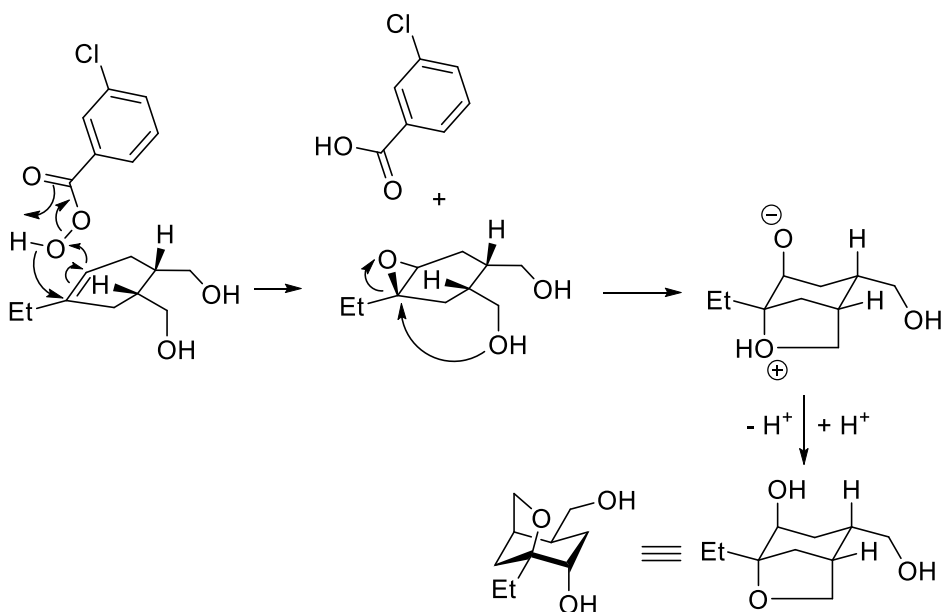
16.2



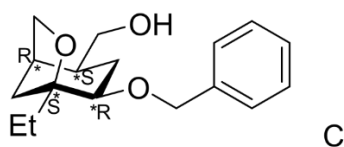
16.3



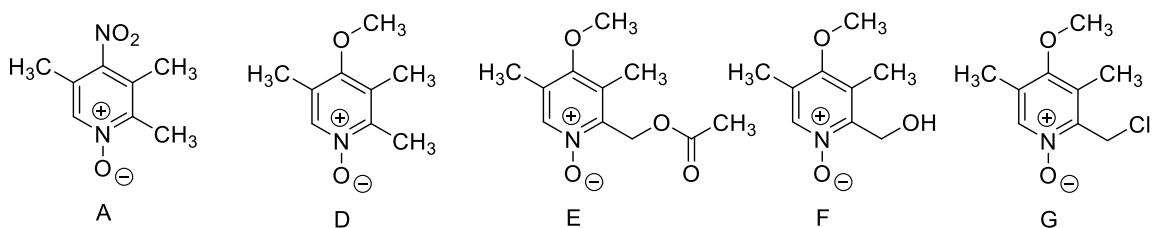
16.4



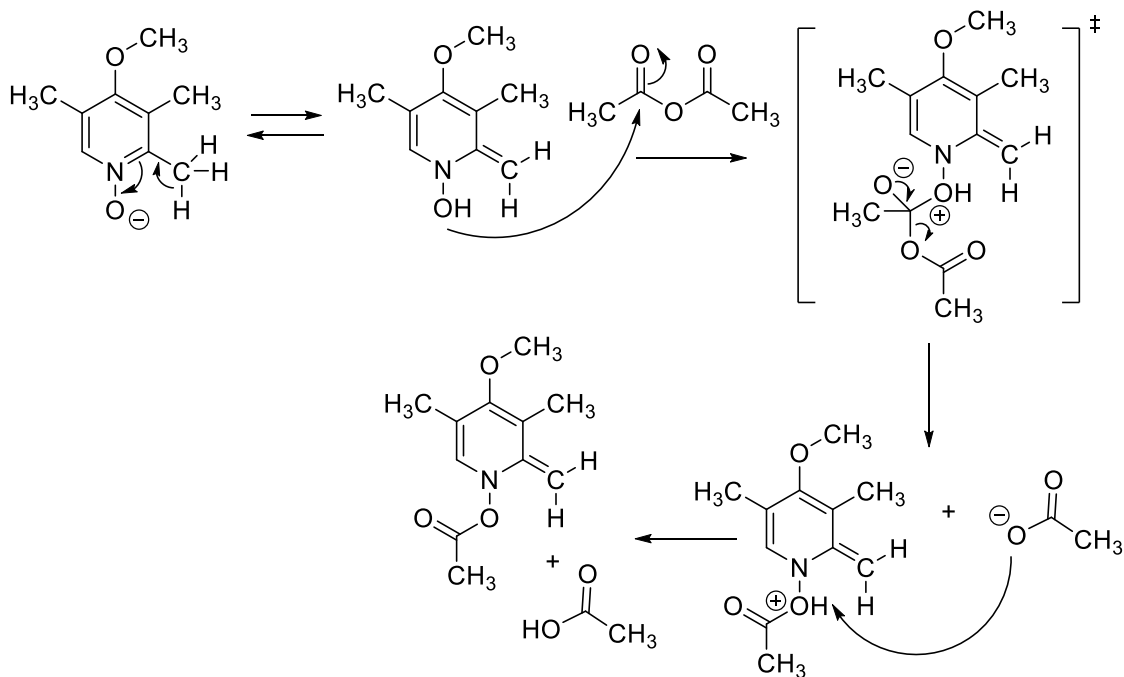
16.5

**Problem 17**

17.1 Starting material is named: 2,3,5-trimethylpyridin-N-oxide

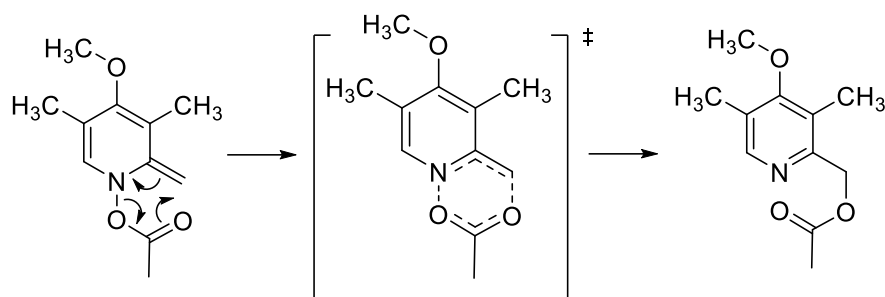


17.2 Step 1: tautomerism



17.3 Reagents B and C are sodium hydroxide and methanol.

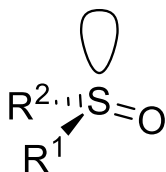
17.4 Mechanism:



The driving force of this reaction is the regeneration of the aromatic system.

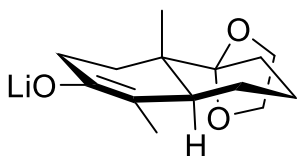
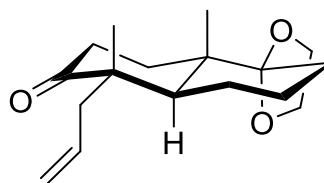
17.5 MCPBA is used to oxidize the thioether obtained in the last step.

17.6 The product of the synthesis is chiral. The stereogenic center is the sulfur atom, which has a non-bonding electron pair and adopts a tetrahedral conformation.



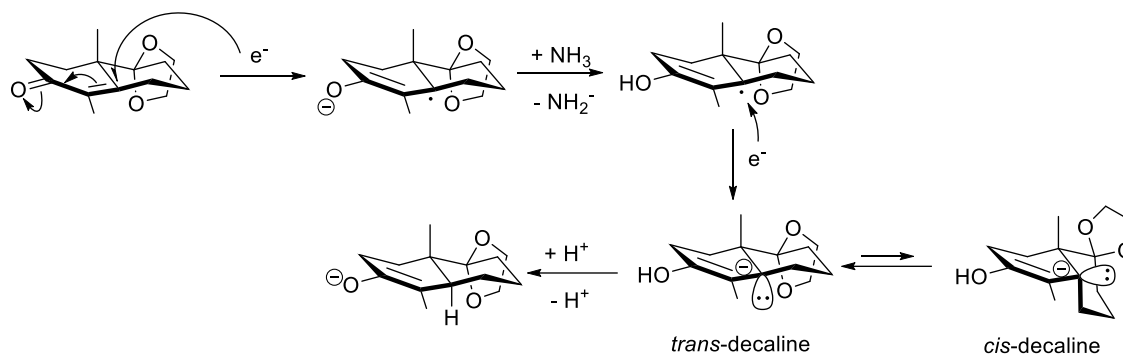
Problem 18

18.1.1

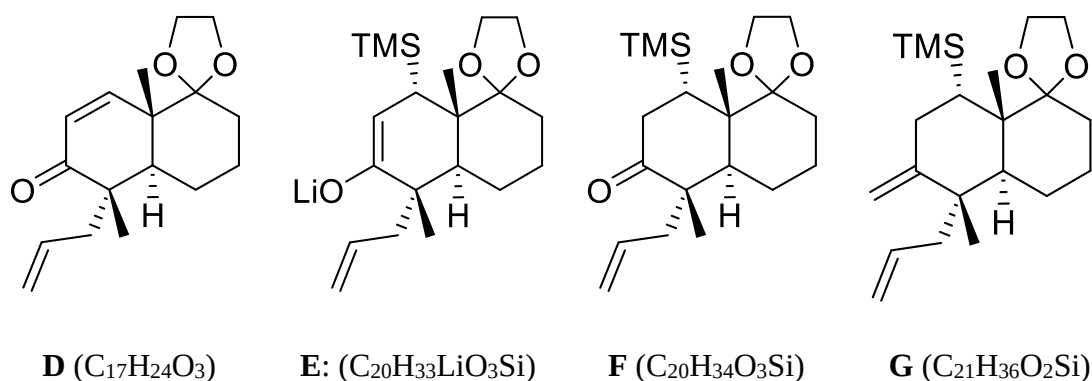
**B** ($\text{C}_{14}\text{H}_{21}\text{LiO}_3$)**C** ($\text{C}_{17}\text{H}_{26}\text{O}_3$)

Reducing reagents of dissolved metals, as thus used in the conditions of Birch's reduction, usually don't reduce isolated double bonds. However, the reaction is possible when the double bond is conjugated because the intermediates are stabilized by electronic delocalization. During the sequence of addition of electrons to the ring and proton transfers that happen, an anionic intermediate is formed with the *trans*-decaline system. This intermediate is in equilibrium with the *cis*-decaline conformation, but in this case the most stable *trans*-decaline conformation predominates, because it makes possible the delocalization of the electrons via the conjugation of the carbanion and the π system of the double bond. As result, the formation of the *trans* diastereoisomer of **B** proceeds selectively.

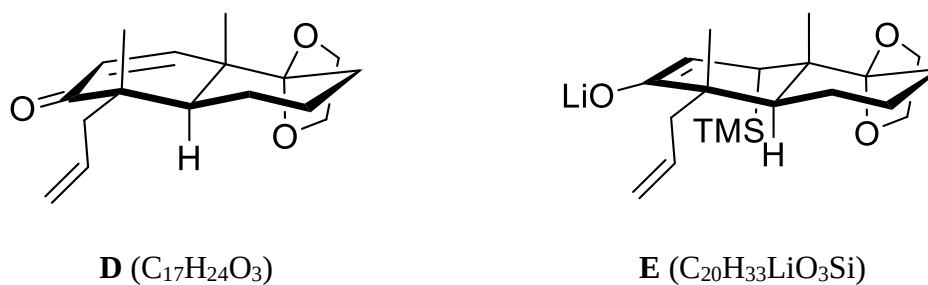
The alkylation of enolate **B** with allyl bromide is diastereoselective because the reaction will occur exclusively by the least hindered face: the face opposed to the axial methyl group in the bridge head.



18.2.1.

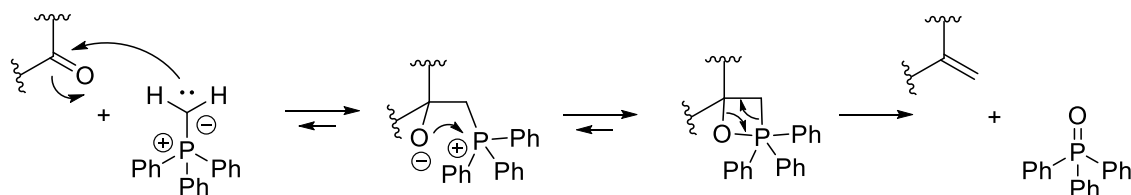


18.2.2.

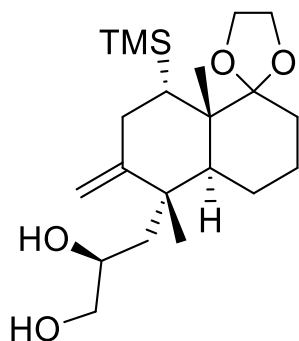
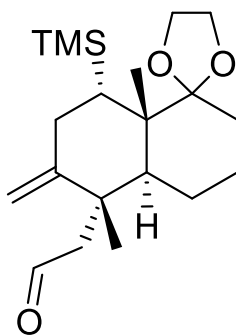
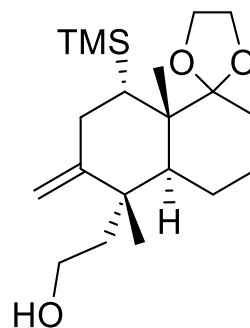


The addition of the trimethylsilyl anion to the α,β -unsaturated ketone **D** is diastereoselective because the reaction occurs exclusively by the least hindered face: the face opposed to the two vicinal axial methyl groups.

18.2.3. Reaction 5 is named Wittig's reaction.

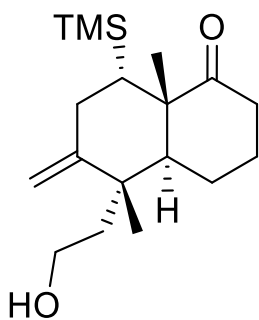
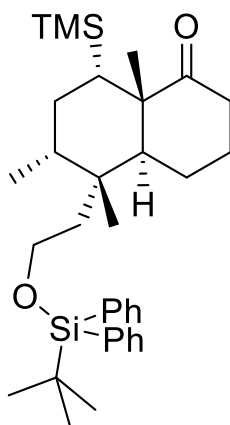
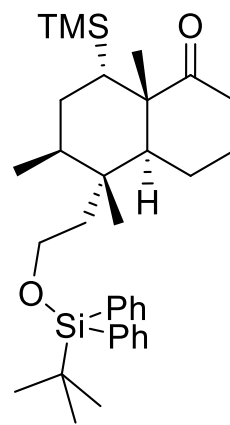


18.3.1.

**H** ($C_{21}H_{38}O_4Si$)**I** ($C_{20}H_{34}O_3Si$)**J** ($C_{20}H_{36}O_3Si$)

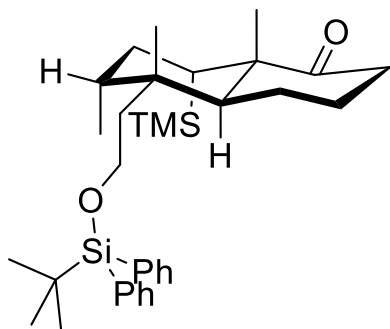
18.3.2. The additional advantage of Sharpless's method is that using of a hindered ligand $(DHQD)_2PYDZ$ increases the selectivity of the hydroxylation towards the monosubstituted alkene of the lateral chain in comparison to the 1,1-disubstituted exocyclic alkene. Thus, the dihydroxylation reactions proceed regioselectively through the vinyl group, which is not possible through traditional methods.

18.4.1.

**K** ($C_{18}H_{32}O_2Si$)**L** ($C_{34}H_{52}O_2Si_2$)**L'** ($C_{34}H_{52}O_2Si_2$)

18.4.2. Reagent b.

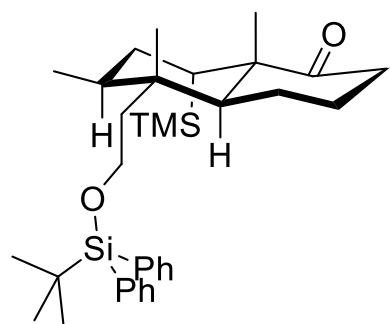
18.4.3.

**L**

$$E_D = U_{Me} + U_{TMS} + \frac{1}{2} (A_{Me} + A_{TMS})$$

$$E_D = 1.8 + 6 + \frac{1}{2} (1.8 + 6)$$

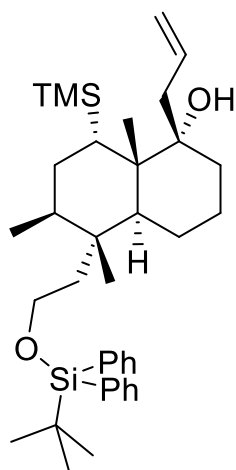
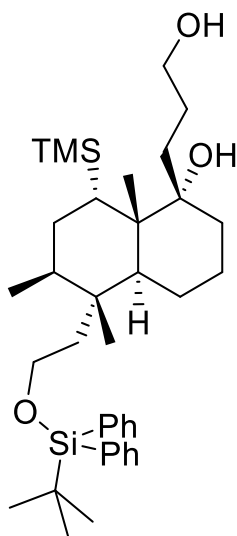
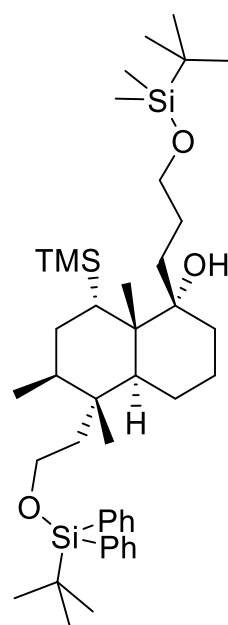
$$E_D = 11,7 \text{ kcal/mol}$$

**L'**

$$ED = A_{TMS} + G_{Me} + G_{Me}$$

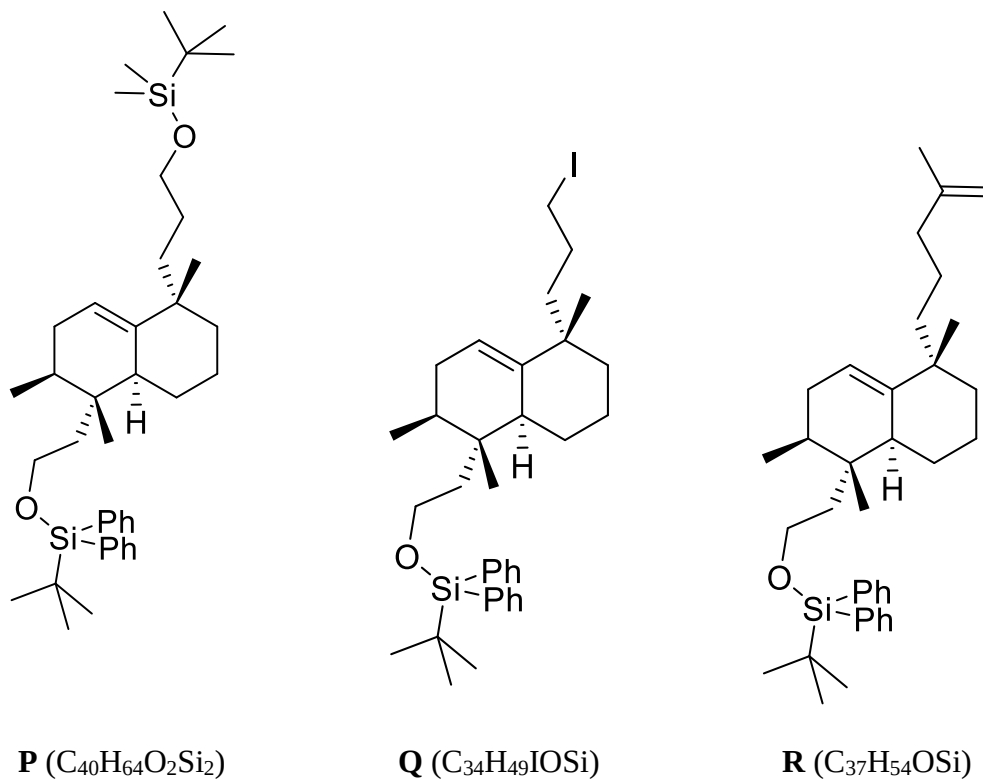
$$ED = 6 + 0,9 + 0,9$$

$$ED = 7,8 \text{ kcal/mol}$$

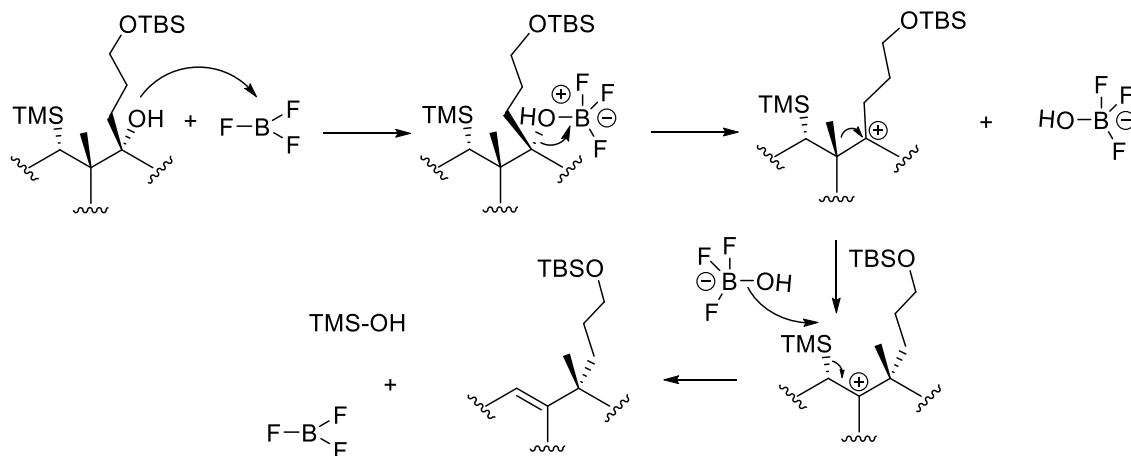
18.5.1.**M** (C₃₇H₅₈O₂Si₂)**N** (C₃₇H₆₀O₃Si₂)**O** (C₄₃H₇₄O₃Si₃)

18.5.2. The nucleophilic addition of allylmagnesium bromide to ketone **L** is stereospecific because the reaction will happen exclusively by the least hindered face: the face opposed to the voluminous group trimethylsilyl in axial position.

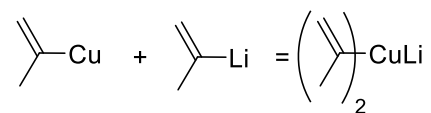
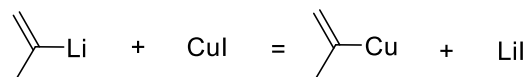
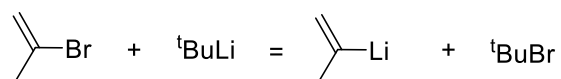
18.6.1.



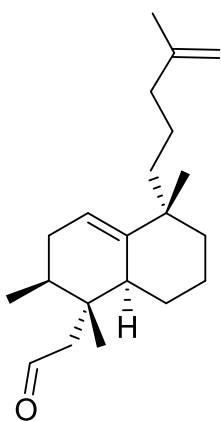
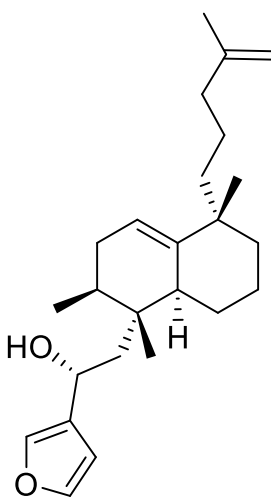
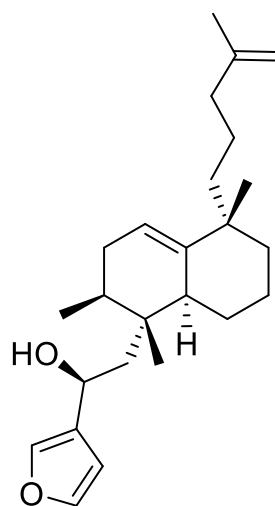
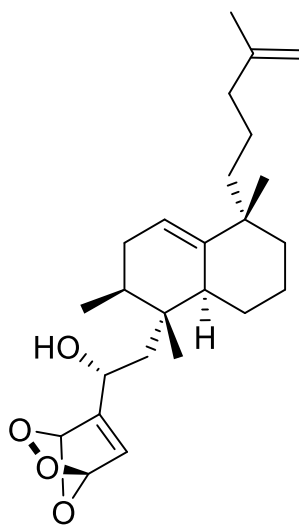
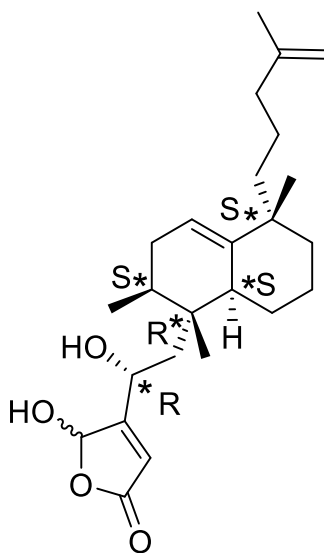
18.6.2. Si-C bonds are larger and weaker than C-C bonds. C-Si bond is slightly polarized to the carbon atom.



18.6.3



18.7.1

S (C₂₁H₃₄O)T (C₂₅H₃₈O₂)T' (C₂₅H₃₈O₂)U (C₂₅H₃₈O₄)

disidiolide

Annex 1. Topics of Organic Chemistry for the National Contest

Fundamentals of structure, bonding and reactivity applied to the organic molecules

Nomenclature of the main families of organic molecules.

Stereochemistry. Elements of molecular chirality, asymmetric center, plane and symmetry axes; atoms that may be asymmetric (C, S, N, P); chirality of a molecule, chirality of allenes and biphenyls; absolute stereochemistry and relative stereochemistry; notation of Cahn-Ingold-Prelog; use of tridimensional representations; E and Z notation for alkenes; kind of stereoisomers, physical and chemical properties, optical activity; racemization and racemic mixture; meso, erythro and threo stereoisomers.

Conformational analysis. Conformations of alkanes and cycloalkanes; torsion tension and sterical hindrance; eclipsed, anti and gauche conformations; envelope conformation of cyclopentane; chair and boat conformations of cyclohexane, axial and equatorial positions; see Newman's and Caballero's projections; cis and trans fusions in cyclic compounds and in fused bicycles; Bredt's rule for bridged bicycles.

Intermediaries in organic reactions: carbocations, carbanions, radicals and carbenes; formation; relative stability.

Stereoelectronic effects. Resonance, conjugation and hyperconjugation; inductive effect; mesomeric effect; aromaticity and antiaromaticity, Hückel's rule; electron rich π rings and electron deficient π rings; neighboring group effect (anchimeric assistance).

Reactivity. Nucleophiles and electrophiles; leaving groups; acidity and basicity of organic molecules, determinant factors; comparison between S_N1 , S_N2 , $E1$ and $E2$, factors that favor a particular mechanism (electronic effects, sterical hindrance, strength of the nucleophile/base, solvent and temperature, kinetics, transposition); quantitative calculation of the regioselectivity of the halogenation, difference between reactivity and selectivity, Hammond postulate; Markovnikov's rule for the electrophilic addition of alkenes and alkynes, peroxide effect or Kharasch-Mayo effect in the addition of HBr to alkenes; Pearson's theory of reactivity or hard and soft acid base theory, Michael's addition vs 1,2 addition, S_N2 and S_N2' reactions in allylic systems; causes of the distinct reactivity of alkenes, alkynes, dienes and aromatic derivatives; kinetic control and thermodynamic control in the reactions of conjugated dienes (1,2 and 1,4 products) and in the Diels-Alder's reaction (endo and exo products); relative stability of the σ complex in S_NAr and S_EAr reactions, activants

and deactivant groups, orientation of the reactions in substituted derivatives of benzene, furane, pyrrole, thiophane, pyridine, naphthalene, anthracene and phenanthrene; regioselectivity and stereoselectivity of the chemical reactions; relative reactivity of the carbonylic compounds towards nucleophiles, influence of steric and electronic effects and of the leaving group.

Radical reactions

Alkane halogenation; Kolbe's synthesis of alkanes; allylic and benzylic halogenation; Hunsdiecker's reaction; alkane pyrolysis and combustion reaction.

Organometallic reagents

Oxidative addition of Mg or Li in alkyl, aryl and vinyl halides; alkyne deprotonation; reactions with carbon dioxide, formaldehyde, aldehyde and ketones; transmetalation reactions to form other organometallic reagents (Cu, Zn, Cd); coupling reactions between organometallic reagents and halides (Corey-House's synthesis, Wurtz's reaction, Ullmann's reaction).

Nucleophilic substitution in the saturated carbon

S_N1 and S_N2 reactions of halides, sulfonates (tosylates, mesylates, triflates) and similar; reactions with water, silver salts, thiols, alcohols, phosphines, amines, azides, cyanides, alkynes, Gilman's reagents, phthalimides (Gabriel's synthesis), phenols (Williamson's synthesis) and thiophenols, halides (Fittig's reaction), (thio)carboxylic acids; reactions of alcohols with hydrazides; exhaustive methylation with Me₂SO₄ and MeI; methylation of phenols and carboxylic acids with diazomethane; reaction of alcohols with phosphorous halides; deprotection of phenolic ethers with BBr₃; epoxide opening in basic and acid media with water, alcohols, amines, cyanides, hydrides, azide; Ritter's reaction; epoxide formation from halohydrins; reaction of ethers with hydrogen halides and water in acid media.

Elimination reaction

E1 and E2 reaction with different kind of bases and leaving groups; Hofmann's and Zaitsev's rules; Hofmann's elimination; Cope's elimination; eliminations in carbocations; dehydration of alcohols; alkyne formation from vicinal dihalides and vinyl halides, regioselectivity; conjugate elimination (E_{1c}B) in the products of aldolic addition and similar.

Electrophilic addition to alkenes and alkynes

Reaction of alkenes with halogens and similar (IBr, ClSR), peroxides (Prilezhaev's reaction), hydrogen halide, N-halosuccinimides, sulfuric acid, boranes; Simmons-Smith's reaction; ozonolysis (oxidative, reductive and neutral), halohydrins formation and similar; oxy(alkoxy)mercuration demercuration; electrophilic addition to dienes at low and high temperature; hydration of alkenes and alkynes; polymerization of acetylene.

Addition to electrophilic alkenes

Reaction of conjugated double bonds with electronwithdrawing groups (amides, esters, ketones, aldehydes, acyl chlorides, nitro...) with cyanhydric acid, Grignard's reagent, Gilman's reagent, organolithiums, ammonia, amines, sodium borohydride, lithium and aluminum hydride, thiols, alcohols, phenols, thiophenols, carboxylic acid salts, phosphines, peroxides, hydrogen halides, acetylides and enolates (Michael's reaction).

Nucleophilic addition to the carbonyl group

Formation of cyanohydrins; reaction of aldehydes and ketones with hydrides, organometallic reagents, water, alcohols and bisulfite.

Nucleophilic substitution in the carbonyl group with loss of an oxygen atom

Formation of linear and cyclic acetals and ketals; formation of 1,3-dithianes; reactions of aldehydes and ketones with amines, hydroxylamine, hydrazine, (2,4-dinitro)phenylhydrazine and semicarbazide; reaction with POCl_3 and PCl_5 ; reductive amination; Strecker's synthesis; Wittig's reaction.

Nucleophilic substitution at the carbonyl group

Relative reactivity of acyl chlorides, anhydrides, esters, amides and carboxylic acids; reactions with amines, thiols, alcohols, water and carboxylic acids in acid or basic media; reaction of carboxylic acids with SOCl_2 , PCl_5 and oxalyl chlorides; partial and total hydrolysis of nitriles; reaction with organolithiums, Grignard's and Gilman's reagent; reduction with lithium aluminum hydride and diborane; formation of sulfonamides and tosylates; addition of amines and alcohols to isocyanates; reaction of diazomethane with acid halides.

Reactions of enols, enolates and similar

Alkylation and acylation of enamides; formation of enols and enolates of esters, ketones, aldehydes, acid halides, nitriles, sulfones, sulfoxides and nitro compounds; reaction of enols and enolates with alkyl halides, acyl halides, sulfonic acid esters, halogens and their equivalents (NIS), epoxides, nitrous acid; haloform reaction; reactions of substitution of 1,3-dithianes; aldolic addition and condensation; Claisen's and Dieckmann's reaction; malonic synthesis, decarboxylation of malonic acids; acetoacetic synthesis, decarboxylation of α -ketoacids and of α -ketoesters (Krapcho's reaction); Hell-Volhard-Zelinsky's reaction; analogues of aldolic and Claisen's reactions in compounds with active hydrogens (modifications of Knoevenagel, Doebner, Perkin, nitroaldolic or Henry, Reformatsky, Stobbe, Mannich, Darzens).

Electrophilic aromatic substitution

Halogenation, nitration, sulfonation of benzene and their derivatives; Friedel-Crafts's alkylation and acylation; Kolb -Schmidt's carbonatation; reactions of Gattermann and Gattermann-Koch; Reimer-Tiemann's reaction; Hoesch's reaction; chloromethylation; nitrosation.

Nucleophilic aromatic substitution

Reaction of deactivated halobenzenes with amines, alcohols, thiols, water, hydrazine, cyanides, phenols, thiophenols and azide.

Cycloaddition reactions

Diels-Alder's reaction; alkene hydroxylation of OsO₄; ozonolysis; insertion of carbenes to alkenes, Simmons-Smith's reaction.

Rearrangements and fragmentations

Picolinic and benzylic acid rearrangements; reactions of Baeyer-Villiger and Dakin; migrations in carbocations; Beckmann's transposition; Hofmann's transposition; Tischenko's transposition.

Oxidations and reductions

Reactions of reduction with LiAlH₄, NaBH₄, H₂/catalyzer, B₂H₆, N₂H₂, NaBH₃CN, DIBALH, metal/NH₃(l); reduction of Birch, Zn(Hg)/HCl (reduction of Clemmensen), Sn/HCl, NH₂NH₂/base (reaction of Wolf-Kirchner), Na/alcohol; obtention of pinacols.

Oxidation reactions with KMnO_4 , HIO_4 , H_2CrO_4 , PCC, PDC, H_2O_2 , AMCPB, O_3 , OsO_4 . Cannizaro's reaction; Meerwein-Ponndorf-Verley's reduction and Oppenauer's oxidation; allylic oxidation with CrO_3 and SeO_2 ; Riley's oxidation.

General mechanism for the oxidation of alcohols and aldehydes.

Other reactions

Amide and oxime dehydration; addition of Grignard's reagents to nitriles; formation of diazonium salts; reaction of diazonium salts with CuX , KI, water, H_3PO_2 , anilines, phenols, malonic ester and acetoacetic ester; installation and removal of protecting groups (thio)acetal, THP, Bn, Me, Et, tBu, TMS, Boc, Cbz; formation of sodium, copper and silver acetylides; Arndt-Eistert's reaction; Nef's reaction; borane oxidation with peroxide to give alcohols.

Identification of organic molecules

Hinsberg's test; haloform reaction; Lucas's test; reaction with Na; reaction with bromine water; Baeyer's reagent; Tollens's reagent; Fehling's reagent; Benedict's reagent; reaction with AgNO_3 ; reaction with FeCl_3 ; reaction with HCl; reaction with NaOH; reaction with Na_2CO_3 ; reaction with NaHCO_3 ; Jones' reagent; reaction with the aqueous solution of NaI and NaIO_3 ; alkaline fusion with Na followed by reaction with AgNO_3 , PbAc_2 or $\text{Na}_2[\text{Fe}(\text{CN})_5\text{NOS}]$ and FeSO_4 .